

Communicative Efficiency in Adjective Ordering: Plan-Guided Production between Hierarchical Composition and Incremental Pragmatics

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Abstract

Adjective ordering must reconcile stable order expectations with the usefulness of properties in the current scene. Theoretical accounts locate communicative efficiency in two directions: hierarchical semantic composition interprets noun-adjacent modifiers before outer modifiers, whereas surface production makes linearly earlier information available first. We test how these pressures are expressed in explicit order judgements and in the full distribution of produced utterances. A reanalysis of German slider ratings and a direct replication show a context gradient and a small residual size-first preference when colour alone identifies the target. A production study shows that speakers often resolve reference with one adjective, so adjective selection and utterance length can pre-empt an ordering decision. We compare matched Rational Speech Act models over fifteen completed responses. After controlling participant-specific variation in stable order expectations, plan-guided incremental production predicts responses better than global and fully incremental alternatives. Its predictive gains are concentrated in colour-initial responses, and participant-specific weights further improve prediction of multi-adjective utterances. Context-fixed semantics predicts the production responses more accurately than sequential context updating, with the difference concentrated in adjective content and length. Controlled simulations show that a size-first target-recovery advantage is compatible with both semantic regimes and depends on prefix evaluation, the utterance space, and semantic parameters. Together, the results connect adjective ordering to the joint evaluation of what to mention, how much to say, and which alternatives compete during production.

Keywords: adjective ordering, reference production, Rational Speech Act, incremental production, Bayesian cognitive modelling

1 Introduction

Adjective ordering provides a window onto how stable order preferences interact with scene-specific informativeness in referential communication. In languages with prenominal modification, robust ordering preferences favour size-first sequences such as *the big blue sticker over the blue big sticker* (Sproat and Shih, 1991; Gregory Scontras, 2023). Yet the first adjective is also the listener’s first descriptive cue to the target. In a scene where colour alone identifies the intended sticker, *blue* would provide the decisive information sooner, even though a colour-first sequence conflicts with the canonical size-first preference (Fukumura, 2018; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021). The order that sounds most natural may therefore differ from the order that guides the listener most efficiently through the scene.

What counts as efficient—and therefore early—depends on the direction in which an adjective string is evaluated. One line of work concerns adjective subjectivity. More subjective adjectives tend to precede less subjective adjectives in prenominal sequences (Gregory Scontras, Degen, and Noah D Goodman, 2017). Under hierarchical semantic composition, the adjective next to the noun combines with the noun before an adjective farther away. In *the big blue sticker*, *blue* therefore combines with *sticker* before the gradable adjective *big* is interpreted. Because colour is typically less subjective than size, this restriction can provide a more stable comparison class for interpreting size and thereby contribute to a size-first surface preference (Franke, Gregory Scontras, and Simon, 2019; Gregory Scontras, Degen, and Noah D Goodman, 2019; Gregory Scontras, Bar-Sever, et al., 2020).

In reference games (Franke, 2009; Frank and Noah D Goodman, 2012), communicative efficiency concerns how an utterance guides the listener to the target as its words unfold. Referential usefulness therefore assigns priority through the surface linear sequence. Listeners encounter the first spoken adjective before the rest of the utterance and can use it to begin restricting the set of possible referents (Tanenhaus et al., 1995; Sedivy et al., 1999; Rubio-Fernandez and Jara-Ettinger, 2020). When colour distinguishes the target more effectively than size, placing colour first can make the decisive information available sooner (Fukumura, 2018; Rubio-Fernández, Mollica,

and Jara-Ettinger, 2021). The same size–colour string is therefore traversed in two directions. Hierarchical composition reaches colour before size. Surface linear ordering reaches size before colour. Communicative efficiency can favour different surface orders depending on whether it operates through hierarchical semantic composition or surface linear ordering.

Moreover, before a speaker can weigh alternative adjective orders, they first decide which properties to mention and when to stop. In a colour-sufficient scene, *the blue sticker* may complete the utterance, so competition between size–colour and colour–size orders may never arise. In a size-sufficient scene, speakers may still produce *the big blue sticker*. This creates an asymmetry in the redundant use of colour and size adjectives. Previous empirical studies show that colour is included redundantly more often than relative adjectives such as size (Tarenskeen, Broersma, and Geurts, 2015; Rubio-Fernández, 2016). Such redundancy can still support efficient identification, especially when modifier meanings are uncertain and gradable, as with size adjectives (Rubio-Fernández, 2016; Degen et al., 2020). Scene-specific usefulness may therefore affect which adjectives are selected and when production stops, as well as how jointly selected adjectives are ordered.

These choices make the full distribution of one-, two-, and three-adjective utterances the relevant production outcome. Within this response space, adjective order is conditional on adjective selection and utterance length. This broader response space also raises a question about production architecture. A globally normalised speaker compares complete utterances as wholes. A fully incremental speaker evaluates each next adjective against the alternatives available after the words already produced. Between these endpoints, an utterance-level plan may constrain production while local alternatives continue to influence successive word choices. We call this intermediate hypothesis *plan-guided incremental production*. Under this hypothesis, an utterance-level evaluation combines with the scene-specific usefulness of successive adjective choices. This possibility connects to broader literature on bounded planning, under which speakers can begin speaking before every part of an utterance has been prepared and can continue planning during articulation (Ferreira and Swets, 2002; Konopka and Meyer, 2014). It also shares with good-enough processing the broader idea that language users can rely on representations whose detail is sufficient for the current task (Ferreira, K. G. D. Bailey, and Ferraro, 2002; Frances, 2024). The plan-guided speaker translates this hypothesis into a distributional question by estimating how strongly successive adjective choices affect the probability of each completed utterance.

We therefore ask three questions, moving from the empirical patterns to computational accounts of the mechanisms that may generate them. First, how do stable size-first preferences and scene-specific referential usefulness jointly shape explicit order judgements and the production outcomes of adjective selection, utterance length, and conditional order? Second, after stable order expectations are controlled, does the production distribution favour global evaluation of complete utterances, fully incremental adjective choice, or plan-guided incremental production between these endpoints? Third, does sequential context updating within hierarchical semantic composition improve prediction in the present production data, and how do semantic composition and speaker evaluation jointly determine simulated ordering advantages?

Different mechanisms can generate the same surface distribution, making their separate contributions difficult to identify. We therefore combine carefully controlled behavioural experiments, computational cognitive modelling, simulations across a broader range of randomly generated contexts, and Bayesian predictive model comparison. The empirical programme separates explicit ordering preferences from the production choices that make an order observable. Study 1a reanalyses the slider data of Schlotterbeck and Wang (2023); Study 1b directly replicates the task with a new sample. Both measure explicit preferences between size-first and colour-first referential expressions across referential contexts. Study 2 uses direct production to elicit utterances and records adjective selection, utterance length, and order conditional on producing the same adjective set. Referential context varies the relative usefulness of size and colour within each scene. Moreover, size discriminability is manipulated between participants, making size a more or less reliable visual cue throughout the experiment. Crossing these factors separates the relative usefulness of size in a particular scene from its perceptual reliability throughout the task. The former directly operationalises discriminatory strength. The latter provides a perceptual-reliability diagnostic relevant to the gradable size meaning invoked by subjectivity-based accounts, with lexical subjectivity held constant.

On the modelling side, we formalise these questions within the Rational Speech Act framework (Frank and Noah D Goodman, 2012; Noah D Goodman and Frank, 2016; Degen, 2023). We extend the qualitative model of Schlotterbeck and Wang (2023) from two fixed adjective orders to the full production distribution. That account combined sequential context updating within hierarchical semantic composition with incremental speaker choice and left their separate predictive contributions unresolved. All production models assign probabilities to the same fifteen ordered one-, two-, and three-adjective responses. They share controls for stable order expectations, visual salience, adjective selection, and utterance length. One comparison asks whether the interpretation of size remains fixed in the original context or is recomputed through sequential context updating within hierarchical semantic composition. A second comparison asks how strongly successive adjective choices contribute to the probability

of a completed response. The global and fully incremental speakers define the two endpoints, and the plan-guided speaker estimates the contribution between them.

Approximate leave-one-out prediction evaluates how accurately these models predict the observed production distribution, with uncertainty in the central contrasts assessed across participants (Vehtari, Gelman, and Gabry, 2017). The fitted comparison asks which components add predictive accuracy in the present materials. A separate simulation compares the semantic regimes on the same randomly generated contexts while varying prefix evaluation, the utterance space, and semantic parameters. It asks when updating generates an ordering asymmetry and whether that asymmetry also arises under context-fixed semantics.

Empirically, the slider studies reveal a replicated context gradient: size-first preferences are strongest when size identifies the target and weakest when colour does. A small residual size-first preference remains when colour alone identifies the target. The production study shows that speakers often resolve reference with one adjective, so adjective selection and stopping frequently determine whether an order contrast arises. Lower size discriminability primarily increases the production of redundant adjective material, especially in size-sufficient scenes. After accounting for variation across participants in stable order preferences, predictive accuracy is highest at an intermediate point on the range from global to fully incremental evaluation. The best predictions therefore combine complete-utterance evaluation with continued influence from successive adjective choices. The largest gains occur among colour-initial responses, and allowing successive-choice weights to vary across participants further improves prediction of multi-adjective utterances. This intermediate architecture gives a formal expression to plan-guided incremental production by combining utterance-level evaluation with local competition. Context-fixed semantics predicts the production responses more accurately than sequential context updating within hierarchical semantic composition. Controlled simulations show that a size-first target-recovery advantage is compatible with both semantic regimes and depends on prefix evaluation, the utterance space, and semantic parameters.

The remainder of the paper is structured as follows. Section 2 introduces the two directional accounts of communicative efficiency and relates them to adjective selection and completed-response modelling. Section 3 reports the behavioural studies. Section 4 defines the formal framework for the complete production distribution. Section 5 introduces the matched comparisons, answers the two computational questions in the fitted production data, and examines the best-predicting model. Section 6 evaluates which computational conditions make an ordering advantage depend on semantic updating, and Section 7 considers the implications and limits of the findings.

2 Sources of Adjective Ordering Preferences

Research on adjective ordering has identified several stable predictors of preferred order, including meaning-based factors such as closeness to the noun and inherentness (Sweet, 1898; Whorf, 1945; Martin, 1969), class-based hierarchies (Quirk et al., 1972; Dixon, 1982; Cinque, 1994; Scott, 2002), and information-theoretic measures (Hahn et al., 2018; Futrell, Dyer, and Greg Scontras, 2020; Dyer et al., 2023). These predictors explain substantial structure, while residual variation remains even for strong individual predictors (for overviews, see Dyer et al., 2023; Gregory Scontras, 2023). Referential context contributes further scene-specific variation in ordering preferences (Fukumura, 2018; Trotzke and Wittenberg, 2019). The present account brings together three lines of work: subjectivity and hierarchical semantic composition, discriminatory strength in reference production, and incremental pragmatic models. Their contributions to the full distribution of one-, two-, and three-adjective utterances remain unresolved.

2.1 Subjectivity, Hierarchical Composition, and Adjective Order

A central empirical generalisation in the adjective-ordering literature is the relation between an adjective’s *subjectivity* and its preferred distance from the noun. Gregory Scontras, Degen, and Noah D Goodman (2017) operationalised subjectivity as susceptibility to faultless disagreement and showed that it explains a large share of variance in English ordering preferences, outperforming earlier predictors such as inherentness and intersectivity. Subsequent work suggests that this relation recurs across languages (Trainin and Shetreet, 2021; Gregory Scontras, 2023). It is directly relevant to size–colour sequences because size adjectives are typically more subjective and gradable than colour adjectives, yielding an expected size-first surface preference.

Gregory Scontras, Degen, and Noah D Goodman (2019) locate the explanatory pressure in restrictive composition. Under hierarchical semantic composition, the adjective closest to the noun restricts the nominal denotation before an outer adjective applies. The outer, more subjective adjective is then interpreted relative to a comparison class established by the inner, less subjective adjective. The inner adjective can therefore provide a comparatively stable basis for interpreting the outer gradable adjective. Franke, Gregory Scontras, and Simoncic (2019) formalise

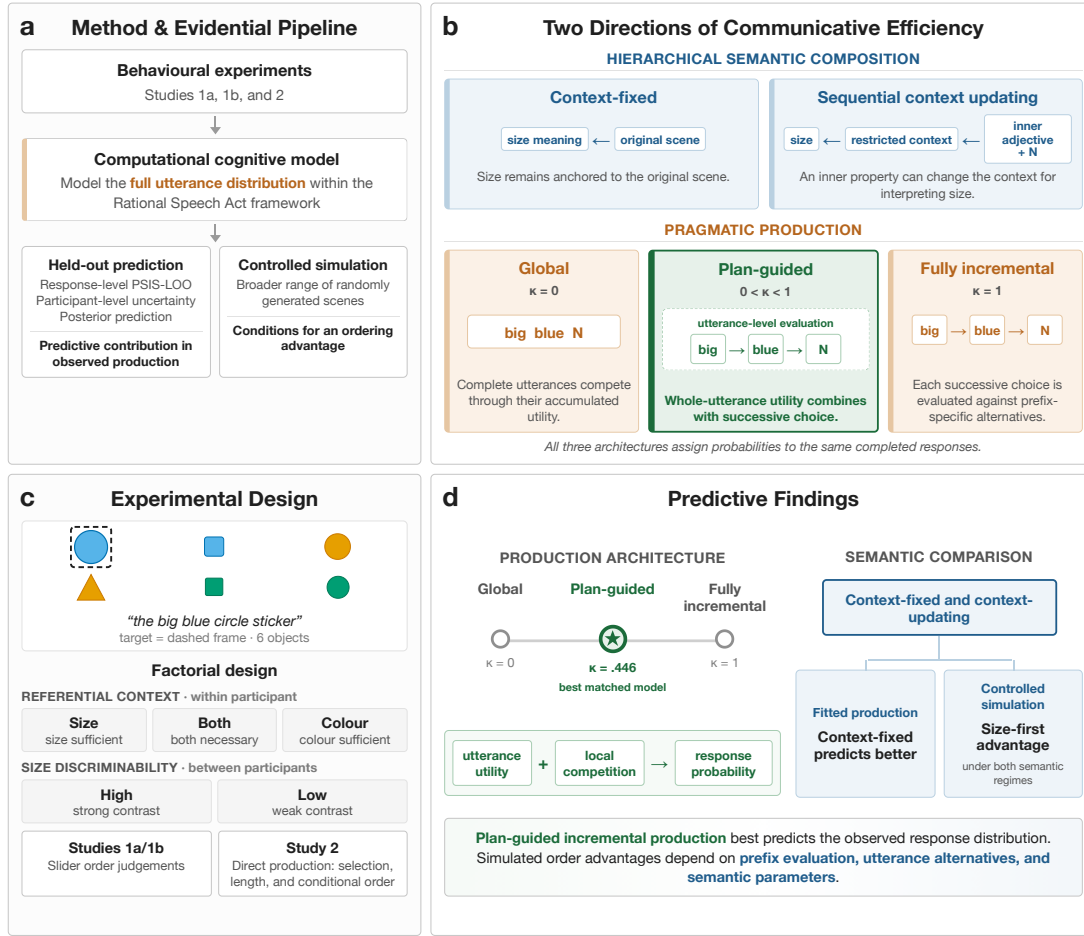


Figure 1: Overview of the design, model space, and evidential structure. Panel A traces the behavioural evidence through computational modelling to held-out prediction and controlled simulation. Panel B distinguishes sequential context updating within hierarchical semantic composition from global, plan-guided, and fully incremental production. Panel C shows the experimental design, and Panel D summarises the predictive result for plan-guided production and the fitted and simulated evidence for sequential context updating.

this intuition in simulation and show that subjectivity-aligned orders can increase average model-assigned target probability under sequential composition with noisy meanings. This account locates the ordering pressure in hierarchical semantic composition. Gregory Scontras, Bar-Sever, et al. (2020) develop the same mechanism as *incremental semantic restriction*. Sequential restriction can therefore generate an ordering pressure when a speaker evaluates complete utterances as wholes.

In the present model, sequential context updating within hierarchical semantic composition implements this proposal. After the inner adjective restricts the relevant context, the interpretation of gradable size can be recomputed relative to that restricted context. Previous simulations establish that this operation can generate an ordering asymmetry under selected semantic assumptions. Its predictive contribution to human production across the full response distribution remains open.

2.2 Referential Usefulness, Adjective Selection, and Length

A second ordering pressure comes from referential usefulness in the surface linear sequence. Fukumura (2018) provide a direct empirical precedent for the present experiments: speakers were more likely to place the more discriminating adjective first. On this account, order reflects which property best distinguishes the target in the current scene. The same adjective can therefore be highly useful in one array and much less useful in another. Rubio-Fernández and colleagues develop this proposal as *incremental communicative efficiency* (Rubio-Fernandez and Jara-Ettinger, 2020; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021). They argue that an early modifier can be useful because it narrows the listener’s candidate set before the utterance is complete. This proposal accords with evidence that listeners integrate partial linguistic input with visual context (Tanenhaus et al., 1995; Sedivy

et al., 1999). From this perspective, word order matters because it determines how the listener’s search space is narrowed as the utterance unfolds.

Referential usefulness also acts before an ordering choice becomes observable. Models of reference planning propose that the contrast set available to the speaker shapes which properties are selected for mention (Fukumura, Van Gompel, and Pickering, 2010; Van Gompel et al., 2019). Perceptual or partner uncertainty can encourage additional, semantically redundant material, and an early redundant modifier can still facilitate visual search (Rubio-Fernández, 2016; Rubio-Fernandez, 2019; Degen et al., 2020). At the maximal end of the response space, producing all three available adjectives has at least two plausible sources. Speaker-internal accounts of reference production, often termed egocentric, propose that perceptually accessible properties can be encoded before the speaker has completed the search for a minimally sufficient description, reducing the planning required to decide which activated property to omit (Pechmann, 1989; Engelhardt, K. G. Bailey, and Ferreira, 2006). Audience-oriented accounts assign the additional material a cooperative role: redundant properties can guide visual search and hedge uncertainty about which cue the listener will use (Rubio-Fernández, 2016; Rubio-Fernandez, 2019; Degen et al., 2020; Tourtouri, Delogu, and Crocker, 2021). In a shared visual scene, these routes can converge because a property that is easy for the speaker to retrieve may also help the listener identify the target. These accounts motivate modelling the pressures on content and length alongside adjective order. A one-adjective utterance can resolve reference without creating an order contrast. Slider ratings hold adjective selection and utterance length fixed by presenting two complete orders. They establish an explicit ordering preference and leave open how often production reaches that choice. The relevant production distribution therefore includes one-, two-, and three-adjective utterances.

2.3 Incremental RSA and the Full Production Distribution

The Rational Speech Act framework formalises rational communication in a shared probability space, allowing semantic interpretation, listener inference, and speaker choice to be combined and compared (Frank and Noah D Goodman, 2012; Noah D Goodman and Frank, 2016; Degen, 2023). Standard RSA models treat complete utterances as the units of speaker choice. The same pragmatic reasoning can be defined incrementally, with the speaker choosing one word at a time relative to the alternatives available after the words already selected (Cohn-Gordon, Noah D. Goodman, and Potts, 2019). Both architectures yield a distribution over completed responses. Variation in planning scope motivates treating global and fully incremental speakers as endpoints for the influence of successive choices (Griffin and Bock, 2000; Ferreira and Swets, 2002; Konopka and Meyer, 2014). Comparing these response distributions tests how strongly successive alternatives contribute to complete-utterance choice. Related models show how incremental pragmatic reasoning can capture asymmetries in modifier position, order, and redundancy (Degen et al., 2020; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021).

Schlotterbeck and Wang (2023) introduced the slider paradigm on which Studies 1a and 1b build. Their qualitative RSA model combined sequential context updating within hierarchical semantic composition with incremental speaker choice. Sequential context updating recomputed the interpretation of gradable size as the relevant context was restricted, while the incremental speaker evaluated successive modifier choices. Together, these mechanisms captured a stable size-first preference and its modulation by discriminatory strength. The analysis considered two fixed size-colour orders and left the separate predictive contributions of the two directions unresolved. The present study extends that starting point with a direct slider replication and a production distribution over one-, two-, and three-adjective utterances, allowing adjective selection, utterance length, and order to be modelled jointly. Within this expanded response space, matched comparisons estimate how much predictive accuracy each direction contributes after stable order expectations and participant variation are controlled. Allowing the contribution of successive adjective choices to vary between the global and fully incremental endpoints provides the distributional test of plan-guided incremental production. A controlled simulation then asks how semantic updating, prefix evaluation, and the available utterances jointly determine an ordering asymmetry.

3 Behavioural Experiments

The three studies expose different aspects of the same empirical problem. Study 1a and its direct replication, Study 1b, ask participants to compare two completed size-colour orders directly using sliders. Study 2 instead elicits utterances through direct production and therefore reveals the richer structure of adjective use in referential communication under our experimental manipulation, including adjective selection, redundant adjective use, utterance length, and ordering conditional on the selected adjectives. The two tasks are complementary: the slider task isolates preferences between two available orders, while the production task situates those preferences within the preceding choices of which adjectives to use and how many to produce. The shared design connects these task formats to the empirical pressures motivating our first research question. Referential context manipulates the

relative communicative efficiency of size and colour within a particular scene by varying whether size, colour, or both are sufficient to identify the target. This manipulation operationalises discriminatory strength, which has been argued to favour placing the more useful property earlier in the surface sequence (Fukumura, 2018; Rubio-Fernandez and Jara-Ettinger, 2020; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021). Size discriminability separately manipulates the overall perceptual reliability of size by varying the size differences within the relevant comparison class. Because size is gradable and interpreted relative to a comparison class, this manipulation provides a diagnostic of whether the reliability of size affects the size-first preference associated with subjectivity and hierarchical semantic composition (Gregory Scontras, Degen, and Noah D Goodman, 2019; Schlotterbeck and Wang, 2023). In direct production, it may also affect whether size is selected and whether speakers add redundant adjective material. Crossing these factors allows us to ask how a stable size-first preference is modulated by scene-specific usefulness and perceptual reliability, both in direct judgements between competing orders and in the production choices on which ordering is conditional.

3.1 General Method Shared between Experiments

3.1.1 Design

The three studies shared the same basic factorial design. The response format differed across experiments. The visual displays and target-definition logic were shared. On every critical trial, participants saw a six-object display with one framed target. The target was always the largest object that matched the intended colour and form. The slider analyses focus on the theoretically central size–colour subset.¹ Within that subset, referential context varied within participants and had three levels: size-sufficient, colour-sufficient, and both-necessary. A second, between-participant factor, size discriminability, varied the size gap between the target and its size competitors. High size-discriminability displays used larger target–competitor gaps, making the size contrast visually pronounced. Low size-discriminability displays used smaller gaps, making size a less reliable cue. This factor was manipulated between participants so that each participant experienced a consistent level of visual size contrast throughout the task (Figure 2).

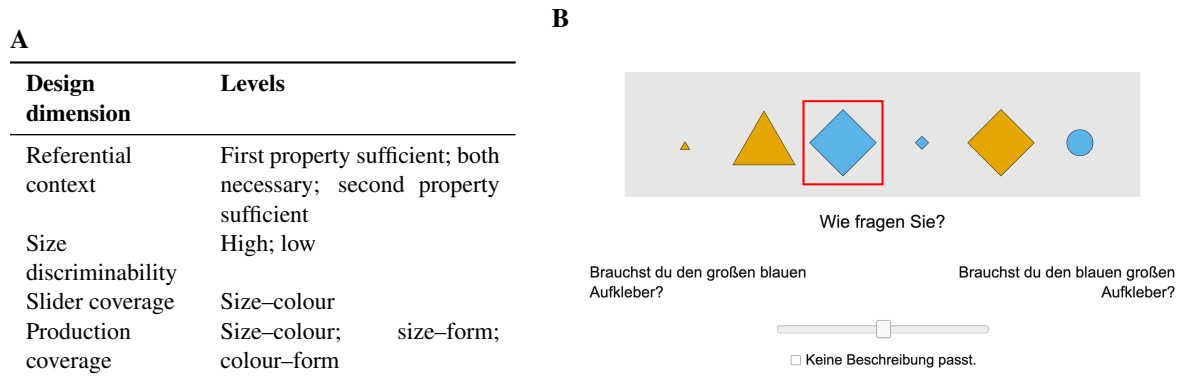


Figure 2: Experimental design and materials. Panel A summarises the shared design dimensions and task coverage. Panel B reproduces an exact slider-trial screenshot: the framed object is the target, and participants rated the two German adjective orders.

3.1.2 Materials

Items were counterbalanced across six lists, each containing nine critical trials in every adjective-family by referential-context combination. In addition to list counterbalancing, the critical scenes were realised through systematic permutations of colour and form values. This created variability in the surface displays and preserved the referential structure required by each condition. These permutations were applied to 27 critical scene templates, each specifying an abstract six-object configuration with a target, competitors, and distractors. Critical vocabularies were fixed across the study, with six colour terms and six form terms reused throughout the scene

¹Study 2 also included size–form and colour–form trials, which enter the complete production model comparison. The size–colour subset remains the theoretical anchor because it provides the clearest contrast between a gradable adjective and a determinate feature adjective. The appendix reports the recorded size levels and list structure.

set; the modified noun was always *Aufkleber* ('sticker').² The computational analysis reconstructed each display deterministically from the raw stimulus record as a 6×3 matrix, with one row per object and columns for observed size (*s*), colour (*c*), and form (*f*). The referent occupies the first row of this matrix. Participants saw the corresponding visual arrays.

3.1.3 Procedure

All studies were conducted online via Prolific. Participants provided informed consent; no formal ethics approval was obtained. Compensation was set slightly above the platform's median level. Participants in each experiment were instructed to identify the framed target using its perceptual properties. Spatial descriptions such as *left* or *right* were excluded. The task was framed as a simple reference game about collectible stickers: participants saw one framed target and gave or evaluated a description for a partner who saw the same stickers in a different order and lacked the target frame. This made spatial terms uninformative and kept the task focused on adjective descriptions. Each study began with a consent form and a brief practice block (three training trials with feedback) that familiarised participants with the interface and the referential task. The main trials were divided into four blocks of approximately equal length, with self-paced breaks between blocks. Trial order within each block was randomised independently for each participant. The experiment-specific response formats, trial counts, and response coding are reported below.

3.2 Study 1a: Slider-Rating Study

3.2.1 Participants

Participants ($N = 134$ after exclusions; 70 in the LOW size discriminability group, 64 in the HIGH size discriminability group) were recruited online via Prolific and reported German as their native language. Exclusion criteria were: more than three missing trial-list entries, completion times more than two standard deviations from the sample mean, more than three failures on control items, or more than three "none fits" responses on experimental items.³ After exclusions, the analysed dataset contained 10,606 critical observations. The size-colour subset reported in the main empirical and model-comparison figures contained 3,485 observations; the appendix summarises exclusion details. Mean completion time was 27.6 minutes. Study 1a reanalyses the same participant sample reported by Schlotterbeck and Wang (2023), using the exclusions and retained sample documented here.

3.2.2 Procedure

Study 1a implemented the shared referential design as a comparative judgement task. On each critical trial, two competing adjective orders appeared on opposite sides of a continuous slider, together with a "none of the descriptions fits" option. Left-right presentation of the two orders was randomised across trials. Each participant saw one of six lists built from the 27 underlying scenes, yielding 81 critical trials and 99 filler and control trials (180 total). The filler and control items discouraged fixed response strategies and ensured attention to the referential task. Responses were reoriented so that larger values favoured the size-first order and rescaled from 0–100 to 0–1. We then subtracted 0.5, yielding a centred scale from -0.5 to 0.5 with zero as the neutral point.

3.2.3 Results

Figure 3 shows a clear referential-context gradient in size-first ratings. We analysed the centred ratings with a Bayesian Gaussian hierarchical model containing referential context, size discriminability, and their interaction, together with correlated participant intercepts and participant slopes for referential context. Planned hypotheses were evaluated through population-level expected ratings; we report posterior medians, 95% credible intervals, and directional posterior probabilities.

Mean centred ratings, indexing preference for size-first orderings, were highest in size-sufficient displays, lower in both-necessary displays, and lowest in colour-sufficient displays. Mean centred ratings remained above zero in colour-sufficient contexts, yielding no qualitative reversal. The posterior contrast between size-sufficient and both-necessary displays was .094, 95% credible interval $[-.061, .127]$, and the contrast between both-necessary and colour-sufficient displays was .171, $[-.131, .208]$. The posterior probability of this graded context order was

²The form values were presented with German adjectival forms, such as *runden* for round/circle and *quadratischen* for square, so that form could occupy the same prenominal adjective position as colour and size.

³Control items were filler trials with an unambiguous expected response, included to screen for inattentive or non-referential responding. The additional "none fits" option allowed participants to indicate that neither description identified the target; frequent use of this option on experimental items was treated as evidence of a response strategy outside the intended comparative-judgement task.

greater than .999. It estimated a colour-sufficient preference of .054, 95% credible interval [.003, .109], with posterior probability .982 of exceeding the neutral point.

The high-minus-low discriminability contrast was .042, 95% credible interval [−.013, .099], in size-sufficient displays, −.043, [−.118, .028], in both-necessary displays, and −.021, [−.129, .082], in colour-sufficient displays. The interaction contrast comparing the high-minus-low effect in both-necessary with size-sufficient displays was −.085, 95% credible interval [−.151, −.022], with posterior probability .995 of being below zero. The corresponding colour-sufficient versus size-sufficient interaction contrast was −.063, 95% credible interval [−.172, .039], with posterior probability .885 of being below zero. The descriptive crossover is therefore concentrated in the difference between size-sufficient and both-necessary contexts, while the colour-sufficient component remains uncertain. Study 1b provides the direct test of whether this interaction pattern recurs. The central result is therefore a size-first preference graded by referential usefulness: it was strongest when size alone identified the target, weakest when colour alone did, and remained slightly positive in the colour-sufficient condition.

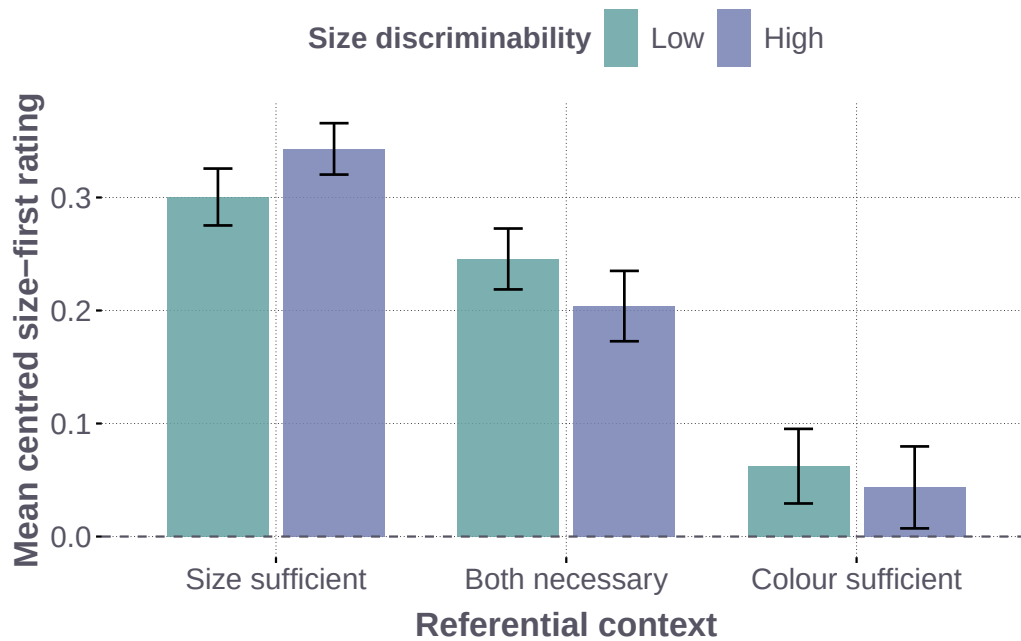


Figure 3: Slider-rating data. Mean centred trial ratings by referential context and size discriminability, with descriptive 95% t intervals; positive values indicate size-first preference.

3.3 Study 1b: Direct Slider Replication

3.3.1 Participants and preprocessing

The replication retained 104 of 132 participants after applying the Study 1a exclusions to an independent participant sample. The cleaned file contains the same 1,080 list-item-condition design tuples as Study 1a and 8,379 critical observations. The size-colour subset contained 2,786 observations. The complete exclusion and preprocessing record appears in Appendix A.

3.3.2 Results

We fitted the same Bayesian Gaussian hierarchical model as in Study 1a to the replication ratings. Mean centred ratings were 0.272 in size-sufficient contexts, 0.197 when both properties were necessary, and 0.054 in colour-sufficient contexts. The posterior contrast between size-sufficient and both-necessary displays was .073, 95% credible interval [.029, .115], and the contrast between both-necessary and colour-sufficient displays was .140, [.090, .190]. The posterior probability of the graded context order was .999. The model estimated the colour-sufficient preference at .057, 95% credible interval [.001, .116], with posterior probability .977 of exceeding neutrality. Study 1b therefore directly replicates the primary context gradient and supplies convergent evidence for a small positive colour-sufficient preference. High discriminability was associated with higher ratings in all three contexts, by .090, 95% credible interval [.014, .160], in size-sufficient displays, .076, [−.011, .161], in

both-necessary displays, and .124, [.016, .240], in colour-sufficient displays. The interaction contrast comparing both-necessary with size-sufficient displays was $-.014$, 95% credible interval $[-.097, .071]$, with posterior probability .623 of being below zero. The corresponding colour-sufficient versus size-sufficient interaction contrast was .034, 95% credible interval $[-.082, .157]$, with posterior probability .715 of exceeding zero. The Study 1a crossover therefore does not recur. Study 1b shows a broadly shared high-discriminability shift across contexts. Size discriminability therefore remains secondary to the replicated context gradient.

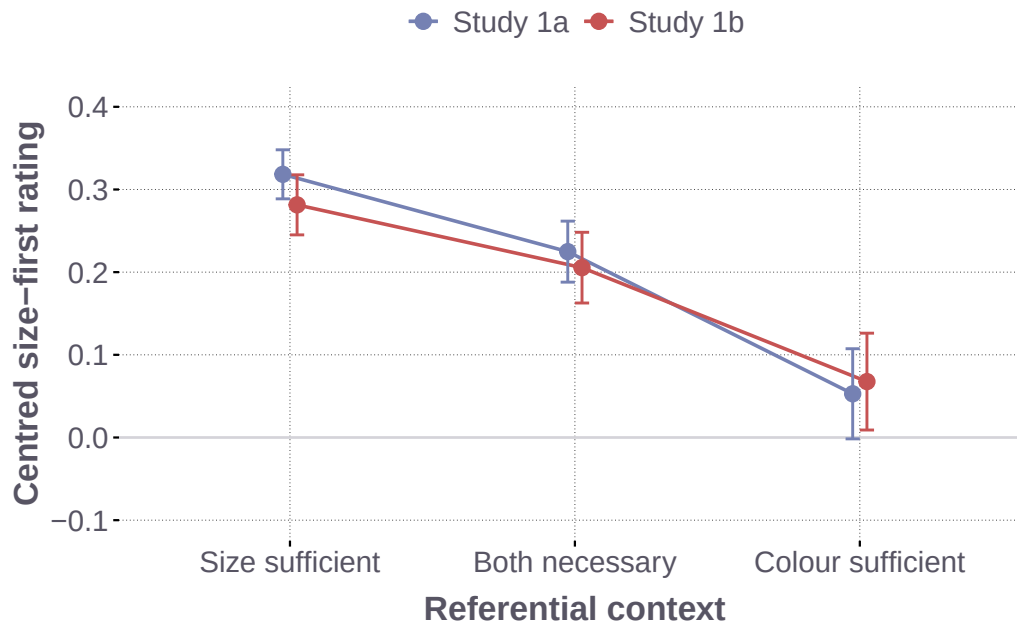


Figure 4: Combined Study 1a and Study 1b slider results. Mean centred ratings by referential context in each study, with 95% confidence intervals over participant-level means; positive values indicate size-first preference.

3.4 Study 2: Production Study

3.4.1 Participants

Participants ($N = 113$ after exclusions; 60 in the LOW size discriminability group, 53 in the HIGH size discriminability group) were recruited online via Prolific; anyone who had completed either slider study was excluded. Further exclusion criteria were: completion times more than two standard deviations from the mean, more than 80% identical responses, or more than 10% missing or unannotatable responses. The identical-response criterion selects a sample that varies its descriptions across trials, which is relevant to the participant variation analysed below. The canonical production table contains 9,100 annotated observations across all three adjective-pair families. The model comparison retains the native fifteen-category response space across size-colour, size-form, and colour-form trials. Adjective family is a structured dimension of the materials that specifies which two properties support reference; the primary experimental factors remain referential context and size discriminability. The targeted Bayesian hypothesis tests use the 3,034 size-colour trials to align directly with the slider studies, while size-form and colour-form broaden the model comparison to the other adjective families in the experiment. Mean completion time was 32.4 minutes. Compensation followed the same Prolific rate as Study 1a.

3.4.2 Procedure

Study 2 used the shared referential design in a production format. Participants saw the framed target together with the fixed sentence frame *den ___ Aufkleber* and typed the adjective material needed before the noun *Aufkleber*. Restricting the response to the adjective slot reduced off-task variability and provided a controlled basis for coding. The practice materials additionally introduced the relevant colour and form vocabulary before the critical trials began. Each participant completed 135 main trials: 81 critical trials and 54 filler or control trials from one of six

counterbalanced lists.⁴

Responses were coded into 15 utterance categories corresponding to all ordered combinations of S (size), C (colour), and F (form): 3 one-adjective, 6 two-adjective, and 6 three-adjective strings. Misspellings and transparent lexical alternatives were normalised when their semantic class was unambiguous; spatial descriptions, comparative or ordinal size expressions, and other unclassifiable responses were excluded. The dependent variable for the Bayesian model comparison is the categorical utterance label $y_i \in \{1, \dots, 15\}$ for each trial i . For supplementary comparison with the slider studies, we also tracked two derived binary projections: whether an utterance was size-initial and whether it contained more adjective material than the minimally sufficient property set. These projections combine several underlying choices and are not treated as direct measures of adjective order. Each projection was analysed with a Bayesian Bernoulli hierarchical model containing referential context, size discriminability, and their interaction, correlated participant intercepts and participant slopes for referential context, and an item intercept. Trials satisfying the coded outcome were coded as 1 and all other annotated size-colour trials were coded as 0. The planned interaction contrasts were evaluated through posterior-predictive response probabilities for a new participant and item.

3.4.3 Results

Figure 5 provides an overview of the ordered response distribution for the 3,034 size-colour trials, which align directly with the slider studies and the targeted behavioural tests. The computational analysis below retains all 9,100 production trials across size-colour, size-form, and colour-form. The complete categorical distribution shows that referential context affects which adjectives are selected and how much adjective material is produced. Both-necessary scenes generally elicit longer descriptions, whereas one-property-sufficient scenes more often elicit a single adjective. In particular, colour-sufficient size-colour scenes are commonly resolved with colour alone. On those trials, no size-colour ordering decision is observed.

Selection, adjective set, and length. In the size-colour trials, speakers preferentially mentioned properties that distinguished the target in the current scene. The resulting unordered adjective-set distribution separates one-property responses from responses that contain both referentially relevant properties and from three-adjective descriptions. Lower size discriminability increased the probability of additional adjective material most clearly when size alone would otherwise have been sufficient. Figure 6A isolates this pattern through the over-informative-response recoding used in the targeted hypothesis test. In the supplementary binary projection, over-informative responses were more frequent under low than high size discriminability in size-sufficient displays (.90 vs. .63). The posterior-predictive low-minus-high difference was .234, 95% credible interval [.151, .318], with posterior probability greater than .999 of exceeding zero. The planned interaction compared this difference with the mean low-minus-high difference in the both-necessary and colour-sufficient contexts; it was .153, 95% credible interval [.072, .236], with posterior probability greater than .999 of exceeding zero. The visual manipulation therefore bears most clearly on adjective selection and utterance length, with weaker evidence for a stable ordering preference.

Initial adjective and conditional order. Initial-adjective probabilities reflect both whether an adjective is selected and where it occurs. For this reason, size-initiality provides a useful descriptive projection of the complete response space whose probability reflects more choices than the slider outcome. Figure 6B shows the size-initial recoding used in the targeted hypothesis test. High discriminability increased size-initial responses in both-necessary displays (.49 vs. .65), with a posterior-predictive difference of .141, 95% credible interval [.036, .242], and posterior probability .997 of exceeding zero. The planned interaction compared this difference with the mean high-minus-low difference in the two one-property-sufficient contexts; it was .169, 95% credible interval [.077, .256], with posterior probability greater than .999 of exceeding zero. The corresponding difference was $-.021$, 95% credible interval $[-.101, .057]$, in size-sufficient displays and $-.032$, $[-.068, -.006]$, in colour-sufficient displays. This interaction concerns the joint outcome of selecting size and placing it initially in the completed response. Conditional on producing both size and colour, size-first descriptions remained more frequent than colour-first descriptions; colour-before-size reversals were comparatively rare. That conditional subset is selected by the prior decision to mention both properties and is therefore reported separately from adjective selection and length. The same distinction applies to size-form, colour-form, and three-adjective responses.

⁴The number of filler and control trials was reduced in Study 2 to keep the total number of trials manageable because typed production responses were slower and more effortful than slider judgements.

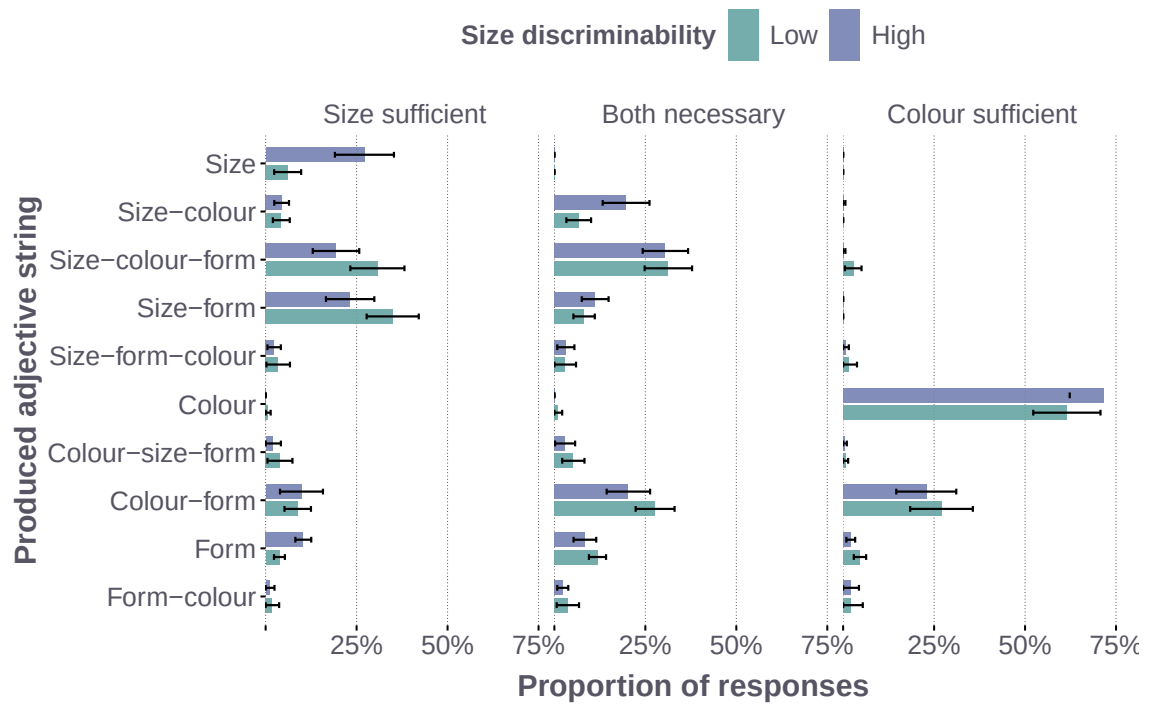


Figure 5: Ordered response distribution across the 3,034 size-colour production trials. Bars show mean participant-level proportions by referential context and size discriminability, with 95% confidence intervals. Adjective labels preserve surface order; the figure shows responses reaching at least 2% in one condition.

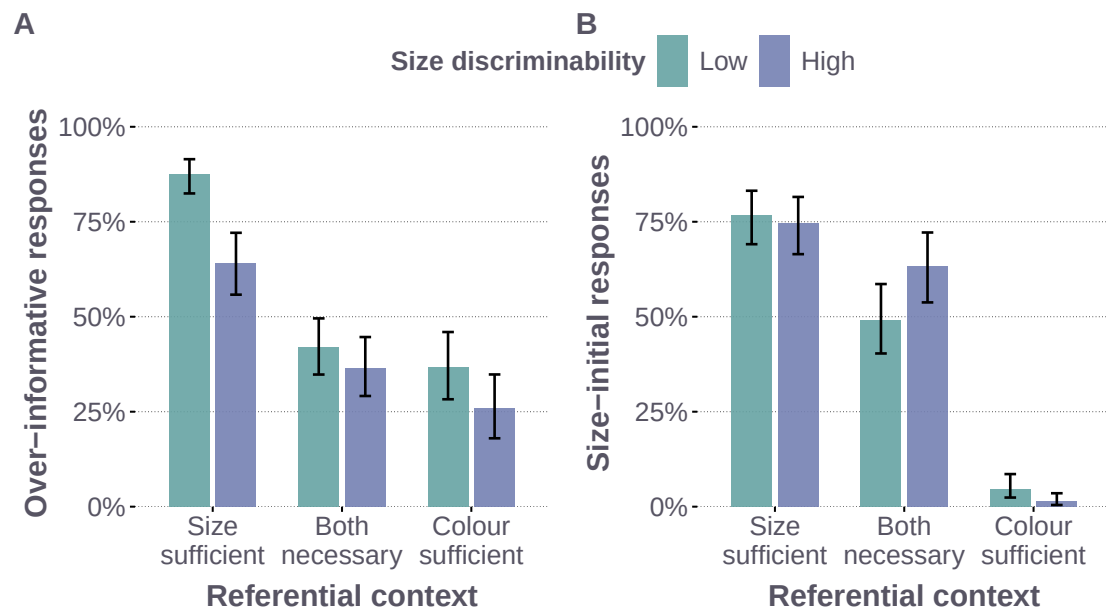


Figure 6: Hypothesis-linked recodings of the size-colour production responses. Panel A codes redundant adjective use as over-informative: a multi-adjective response in a one-property-sufficient context or a three-adjective response when both properties are necessary. Panel B codes whether the response begins with size. Bars show Bayesian posterior medians and error bars show 95% credible intervals. Size-initiality combines adjective selection and order.

3.5 Interim Discussion

The two response formats expose different stages at which communicative efficiency can affect adjective ordering. The slider studies establish a replicated context gradient in explicit comparisons between two completed size–colour orders. Ratings are highest when size identifies the target, lowest when colour does, and retain a small positive size-first estimate in the colour-sufficient condition. Direct production allows the same referential context to affect earlier choices about adjective selection and utterance length. When one property identifies the target, speakers frequently produce a single adjective and stop before an order among multiple adjectives becomes observable. The context manipulation can therefore shift direct judgements between completed orders and also determine whether production reaches an ordering choice at all.

The interaction between size discriminability and referential context differs across tasks and production outcomes. In Study 1a, the interaction is concentrated in the contrast between size-sufficient and both-necessary displays, with weaker evidence for the colour-sufficient component. Study 1b shows a broadly shared high-discriminability shift across contexts, and both interaction contrasts remain uncertain. The slider studies therefore replicate the referential-context gradient while providing limited evidence for a stable discriminability-by-context interaction in explicit order judgements. Study 2 shows stronger outcome-specific interactions within direct production. Lower size discriminability increases additional adjective material most clearly in size-sufficient scenes, whereas higher size discriminability increases size-initial responses most clearly when both size and colour are necessary. These production interactions locate the effect in adjective selection, utterance length, and initial position before order is examined conditional on a fixed adjective set. Their magnitudes are expressed on response-probability scales, while the slider interactions are expressed on the centred-rating scale. The reported production hypothesis tests and Figures 5 and 6 are restricted to the size–colour trials, preserving direct comparability with the slider studies. A descriptive check of the complete production data shows that the between-participant manipulation also changes responses in the colour–form family, where colour and form define the referential contrast. In form-sufficient displays, for example, form-initial responses are more frequent in the high- than in the low-discriminability group (.77 vs. .71). Because discriminability was manipulated between participants, such differences can also reflect group variation in how the task is approached. The production model therefore represents participant variation alongside the visual manipulation, and the controlled simulation isolates the computational conditions under which an ordering asymmetry emerges.

Production also reveals substantial use of maximal three-adjective utterances and pronounced participant heterogeneity in how often these longer utterances occur. Three-adjective responses are theoretically informative because speakers continue beyond the material required for unique reference and traverse multiple adjective positions before completing the utterance. Participant heterogeneity further raises the possibility that the pressures governing adjective selection, length, and successive adjective choice differ across speakers. These qualitative patterns motivate a model of the complete fifteen-category response distribution with participant-level variation. An isolated size-first/colour-first contrast captures only the responses in which both adjectives have already been selected.

4 Formal Modelling Framework

The formal framework of the current paper builds on the central intuition of the Rational Speech Act (RSA) model: rational communication links comprehension and production, such that a listener interprets an utterance to infer the speaker’s intended referent and a pragmatic speaker assigns greater probability to utterances that guide the listener successfully to the target (Frank and Noah D Goodman, 2012). In the vanilla RSA model, the speaker evaluates a fixed set of completed utterances according to their communicative efficiency. The present model extends the vanilla RSA model in three respects. First, it expands the response space to all fifteen ordered one-, two-, and three-adjective utterances available in the experiment, bringing adjective selection, utterance length, and ordering into a single production distribution. This extends the two fixed size–colour orders modelled by Schlotterbeck and Wang (2023) to the richer structure of adjective use observed in direct production. Second, it allows recursive updating of the listener state as adjectival meanings compose from the noun outwards. Third, it extends the speaker to incremental production by allowing adjective choices to be evaluated successively along the surface linear sequence.

The last two extensions represent the two directions through which communicative efficiency may shape adjective ordering. Along the surface linear sequence, each adjective changes the alternatives available at the next position. The production architecture determines how strongly these successive evaluations contribute to the probability of the completed utterance, ranging from global evaluation through plan-guided production to fully incremental adjective choice. Within hierarchical semantic composition, adjectival modification proceeds from the noun outwards. Sequential context updating allows a noun-adjacent adjective to restrict the comparison class

relative to which gradable size is interpreted.

The section develops these components in the following steps. We first define the utterance space, visual state space, and literal listener that are shared by all models. We then formalise global, plan-guided, and fully incremental production as alternative ways of assigning probabilities to completed utterances. Finally, we define how size is interpreted under context-fixed semantics and sequential context updating within hierarchical semantic composition. Additionally, all models include shared components that account for task-specific production patterns in adjective selection and utterance length, together with stable order expectations and perceptual salience. Their complete parameterisation and supporting robustness analyses are documented in the Appendix.

4.1 Utterance Space, State Space, and the Literal Listener

The model assigns probabilities to all fifteen ordered strings formed from size, colour, and form: three one-adjective utterances, six two-adjective utterances, and six three-adjective utterances. This response space allows the model to predict which properties speakers mention, how many adjectives they produce, and how jointly selected adjectives are ordered. To make the calculation concrete, consider a schematic display with a big blue target sticker, a smaller blue sticker, and differently coloured stickers of varying sizes. The available utterances include *big*, *blue*, *big blue*, and *blue big*, together with the form-based and longer alternatives in the complete response space. A small reference game of this kind makes the mapping between utterances and listener states transparent (cf. Degen et al., 2020). All models use the observed six-object display and a Rational Speech Act speaker who favours utterances that guide a listener to the target (Frank and Noah D Goodman, 2012; Noah D Goodman and Frank, 2016; Degen, 2023). Size has a graded meaning defined relative to a comparison class (Schmidt et al., 2009; Franke, Gregory Scontras, and Simonic, 2019). Colour provides a graded target-matching cue; form use enters through salience and the task link, with its lexical reliability fixed at .50. For every complete or partial adjective string, the literal listener L applies adjectival modification from the noun outwards, corresponding to right-to-left composition in the prenominal strings considered here.

Let $u_{\leq t} = w_1 \dots w_t$ be a surface prefix and $P_0(o | X) = 1/6$ the uniform prior over the six objects in display X . Under context-fixed semantics, the literal listener after this prefix is

$$L(o | u_{\leq t}, X) \propto P_0(o | X) \prod_{j=1}^t \llbracket w_j \rrbracket_{\text{fix}}(o). \quad (1)$$

The listener combines the meanings made available by the surface prefix and assigns probability to every object that remains compatible with them. In the running example, *blue* leaves both blue stickers under consideration, whereas adding *big* can concentrate listener probability on the target. This listener provides the target-directed value used by all three production accounts.

4.2 Successive Adjective Choice and Plan-Guided Production

At position t , the utility of candidate adjective a for target o^* is

$$U_t(a | u_{< t}, o^*, X) = \alpha \log L(o^* | u_{< t} + a, X) + \lambda_{\text{salience}} z_a, \quad (2)$$

where α controls sensitivity to listener success and z_a represents the visual salience of the candidate property. Holding perceptual salience constant, if *blue* brings the listener closer to the target than *big*, it receives greater target-directed utility at the first adjective position. For a completed utterance u , the model records both the accumulated utility of its selected adjectives and the competition they faced at successive positions:

$$U^{\text{complete}}(u, o^*, X) = \sum_t U_t(w_t | u_{< t}, o^*, X), \quad (3)$$

$$\mathcal{N}^{\text{local}}(u, o^*, X) = \sum_t \log \sum_{a \in \mathcal{A}(u_{< t})} \exp U_t(a | u_{< t}, o^*, X), \quad (4)$$

where $\mathcal{A}(u_{< t})$ contains the adjective types still available after the current prefix. The complete-utterance utility U^{complete} accumulates the utilities of the adjectives selected along the utterance. The local-normalisation term $\mathcal{N}^{\text{local}}$ accumulates the log denominators of the local softmax choices over adjectives available at each position. Suppressing the shared referent and scene arguments, taking the logarithm of the product of these locally normalised choices shows why this term is subtracted:

$$\begin{aligned} \log \prod_t \frac{\exp U_t(w_t | u_{< t})}{\sum_{a \in \mathcal{A}(u_{< t})} \exp U_t(a | u_{< t})} &= \sum_t U_t(w_t | u_{< t}) - \sum_t \log \sum_{a \in \mathcal{A}(u_{< t})} \exp U_t(a | u_{< t}) \\ &= U^{\text{complete}}(u) - \mathcal{N}^{\text{local}}(u). \end{aligned} \quad (5)$$

A selected adjective therefore contributes less when several alternatives at the same prefix also have high utility. In a simplified two-alternative comparison, if *big* and *blue* have equal utility, each has probability .5, and choosing either contributes $\log(.5) = -\log 2$. If *blue* strongly dominates its alternatives, its local probability approaches one and its log-probability contribution approaches zero. Two completed utterances with similar U^{complete} can therefore receive different probabilities because they pass through prefixes with different degrees of competition.

The probability of a completed utterance is

$$S_\kappa(u \mid o^*, X) \propto \exp\{U^{\text{complete}}(u, o^*, X) - \kappa \mathcal{N}^{\text{local}}(u, o^*, X) + \beta\Omega(u) + T(u, X)\}, \quad (6)$$

The remaining terms capture stable order expectations through $\beta\Omega(u)$ and task-specific pressures on adjective selection and utterance length through $T(u, X)$.⁵ They extend the informativity–cost trade-off of a vanilla RSA speaker: target-directed utility is balanced against conventional order preferences and task-specific production costs. These terms are shared across the three production accounts. The equations express the underlying scores; Appendix B specifies the numerical floors and final response likelihood.

Before the final lapse mixture, the same model can be written at common parameter values as a geometric interpolation between the global and fully incremental endpoints:

$$S_\kappa(u \mid o^*, X) \propto S_G(u \mid o^*, X)^{1-\kappa} S_I(u \mid o^*, X)^\kappa, \quad (7)$$

where S_G and S_I are the endpoint distributions evaluated with the same listener, stable-order component, task link, and parameter values. The proportionality in Equation (7) renormalises the interpolated scores over the fifteen completed utterances.

The parameter κ determines how strongly competition at successive adjective positions contributes to the completed-utterance probability. With $\kappa = 0$, the local normalisers drop out and the *global* speaker normalises over the fifteen completed paths. With $\kappa = 1$, the *fully incremental* speaker retains the complete contribution of locally normalised adjective choices. An interior value defines the *plan-guided* speaker: complete-utterance utility and competition among locally available adjectives jointly determine the response probability. Length choice enters through the comparison of completed responses and $T(u, X)$. Accordingly, κ measures the contribution of successive adjective choice to that distribution.

4.3 Sequential Context Updating within Hierarchical Semantic Composition

The semantic comparison concerns the inner-to-outer direction of hierarchical composition. Let $c_1, \dots, c_T$ denote the modifiers in a completed utterance ordered from the noun outwards, so that $c_1 = w_T$ for pronominal modification. Starting from $C_0 = P_0$, composition successively restricts the context state available to the literal listener:

$$C_j(o) \propto C_{j-1}(o) \llbracket c_j \rrbracket_{\rho_j}(o), \quad j = 1, \dots, T. \quad (8)$$

For *big blue sticker*, this path incorporates *blue* before the gradable meaning of *big* applies. For *blue big sticker*, *big* applies before the colour restriction.

The semantic regime determines the comparison class used for size:

$$\llbracket S \rrbracket_{\rho_j}(o) = \begin{cases} \llbracket S \rrbracket_{\theta_\delta(C)}(o), & \text{context-fixed,} \\ \llbracket S \rrbracket_{\theta_\delta(C_{j-1})}(o), & \text{sequential context updating,} \end{cases} \quad (9)$$

where θ_δ derives a graded size threshold from a comparison class and C is the original scene. Under context-fixed semantics, this threshold remains anchored to the original scene. Under sequential context updating, the threshold is recomputed from the context state left by the hierarchically prior modifier. In the running example, *big blue sticker* can therefore evaluate size among the blue stickers, while *blue big sticker* evaluates size relative to the full scene. When the target stands out more clearly within the blue comparison class, sequential context updating gives the size-first order a referential advantage. The speaker encounters the words in a surface prefix from left to right, while L interprets the available string through noun-outward, right-to-left composition. This comparison asks whether recomputing the size comparison class adds predictive accuracy in the observed production data. The simulation examines how this operation interacts with the evaluation rule and utterance space across a broader range of contexts. The threshold construction and further semantic details appear in Appendices B and G.

⁵The selected components and the status of earlier model-development comparisons are described in Appendices B and D.

5 Model Comparison and Predictive Evidence

Section 4 formalised how hierarchical semantic composition and successive adjective choice redistribute probability across the same fifteen completed utterances as a function of communicative efficiency. Production architecture and semantic regime define a 3×2 comparison: global, plan-guided, and fully incremental production are each paired with context-fixed semantics or sequential context updating. We first compare these six models with matched participant structures, then assess the additional contribution of participant-specific successive-choice weights. Approximate leave-one-out expected log predictive density (ELPD) measures how much probability each model assigns to held-out responses. Posterior-predictive checks show which aspects of the observed distribution the best model captures, and response-level diagnostics connect the predictive differences to adjective choices.

5.1 Inference and Model Comparison Setup

We fit each model to the same 9,100 production trials from 113 participants using Bayesian inference in NumPyro (Phan, Pradhan, and Jankowiak, 2019). Section 4 used unindexed parameters to describe a generic speaker. The fitted models make participant variation explicit: all matched models estimate hierarchies for α_i and β_i , and the final plan-guided refinement also estimates a hierarchy for κ_i . Each likelihood evaluation scores all fifteen completed utterances and performs the required semantic composition and prefix-level normalisation. To make this exact calculation feasible, we precomputed trial-independent utterance and prefix structures, vectorised and compiled the listener and speaker computations in JAX, and ran four NUTS chains on a GPU. Figure 7 separates fixed theoretical choices, estimated parameters, participant variation, and observed responses. The fixed variables A and R index production architecture and semantic regime. For participant i on trial t , the model maps the display and target X_{it} and the stable-order score $\Omega(u)$ to a probability $\pi_{it}(u)$ over the fifteen possible utterances, from which the observed response y_{it} is drawn. The vector λ collects the shared task-link coefficients for adjective selection and utterance length.

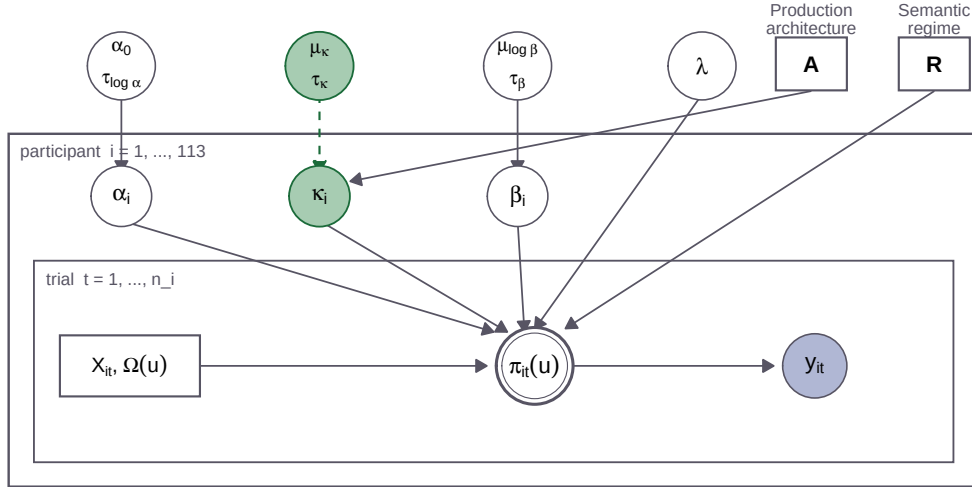


Figure 7: Formal plate for the final joint hierarchical model and the restrictions used in the matched comparisons. Solid circles represent estimated quantities, squares represent fixed model or trial inputs, the double circle represents the deterministic utterance distribution, and the shaded circle represents the observed response. The dashed κ hierarchy is activated only in the final plan-guided refinement.

Higher α_i increases sensitivity to listener success and concentrates probability on higher-utility responses. The parameter κ_i controls how strongly competition among the adjectives available after an emerging prefix contributes to a completed utterance. Its endpoints are global production at $\kappa = 0$ and fully incremental production at $\kappa = 1$. Higher β_i increases the influence of the stable-order score $\Omega(u)$. The task link includes support for a sufficient one-adjective response, task-conditioned form use, visual reliability, and a shared cost for three-adjective utterances. The probability of a three-adjective utterance therefore reflects its accumulated referential value and salience, the shared three-adjective cost, and the participant’s values of α_i , κ_i , and β_i . These quantities

identify how the fitted model distributes maximal utterances while remaining agnostic about the cognitive source of redundant production.

The matched 3×2 comparison provides two architecture-level tests. For production, A selects global, plan-guided, or fully incremental production by fixing $\kappa = 0$, estimating one shared κ , or fixing $\kappa = 1$. The plan-guided-minus-endpoints contrast tests whether allowing an intermediate contribution from successive adjective choices improves prediction. For hierarchical semantic composition, R selects context-fixed semantics or sequential context updating. The updating-minus-fixed contrast tests whether recomputing the interpretation of size after the noun-adjacent adjective improves prediction. All six models share the same response space, task link, semantic parameters, and participant hierarchies for α_i and β_i . These controls isolate the predictive contribution of each proposed direction of communicative efficiency.

The final refinement is confined to plan-guided production. It replaces the shared κ with a population distribution over participant-specific successive-choice weights κ_i and retains the existing hierarchies for α_i and β_i . Its predictive comparison tests whether participant variation in successive adjective choice improves prediction. The population location μ_κ and scale τ_κ then describe where participants lie between the global and fully incremental endpoints and how widely they vary. Section 5.2 reports the matched architecture tests, Section 5.3 evaluates the best-fitted refinement and its absolute adequacy, and Section 5.4 localises its gains across utterances and participants.

Approximate leave-one-out prediction compares out-of-sample accuracy for the complete fifteen-response distribution (Vehtari, Gelman, and Gabry, 2017). The pointwise score vectors use the same trial ordering and response support. Because observations are repeated within participants, paired score contributions are first summed within participant. A participant-level Bayesian bootstrap then assigns Dirichlet weights to the 113 participant contributions over 200,000 draws. Reported uncertainty comprises 95% credible intervals and posterior probabilities for directional contrasts. The predictive target is an additional completed response from a sampled participant, with that participant’s remaining trials available to estimate their participant-specific parameters. The same predictive target is used throughout the architecture and hierarchy comparisons.

5.2 Model-Comparison Results

The matched comparisons quantify the predictive contribution of each proposed direction of communicative efficiency to completed utterances.

Production architecture. Across semantic regimes, the plan-guided model gained 256.9 ELPD over the global model (95% credible interval [71.8, 404.7]) and 548.5 over the fully incremental model ([349.8, 795.1]). Its advantage over the average endpoint was 402.677 ELPD, [311.466, 503.098], with posterior probability above zero greater than .999. Its marginal predictive accuracy was the highest of the three architectures in .994 of the Bayesian-bootstrap draws. The endpoint contrast favoured global production: fully incremental minus global production was -291.596 ELPD, with a wide 95% credible interval of $[-684.673, -1.321]$ and posterior probability above zero of .024. Within the matched plan-guided model, the shared successive-choice weight was

$$\kappa = .446, \quad 95\% \text{ credible interval} = [.407, .486]. \quad (10)$$

The matched comparison supports a joint contribution of complete-utterance evaluation and competition among successive adjective choices.

Hierarchical semantic composition. Updating minus fixed semantics averaged -43.527 ELPD across the three production architectures, with a 95% credible interval of $[-56.005, -31.770]$. The architecture-specific effects all favoured context-fixed semantics: -45.104 ELPD for global production, -61.683 for plan-guided production, and -23.793 for fully incremental production. Each architecture-specific 95% credible interval excluded zero. The implemented updating operation therefore reduces predictive accuracy for the complete response distribution in these materials. The response decomposition below distinguishes its contributions to length, adjective content, and conditional order; Section 6 examines the relation to simulated target recovery.

5.3 Predictive Adequacy of the Best-Fitted Model

The matched comparison identifies the production-architecture advantage with equal participant structure across models. The final refinement tests whether the contribution of successive adjective choices varies across participants within the plan-guided architecture. Adding a participant hierarchy for κ_i gained 364.742 ELPD under context-fixed semantics, with a 95% credible interval of [248.178, 520.967], and 372.285 ELPD under sequential context updating, with a 95% credible interval of [255.597, 526.056]. Both posterior probabilities above zero

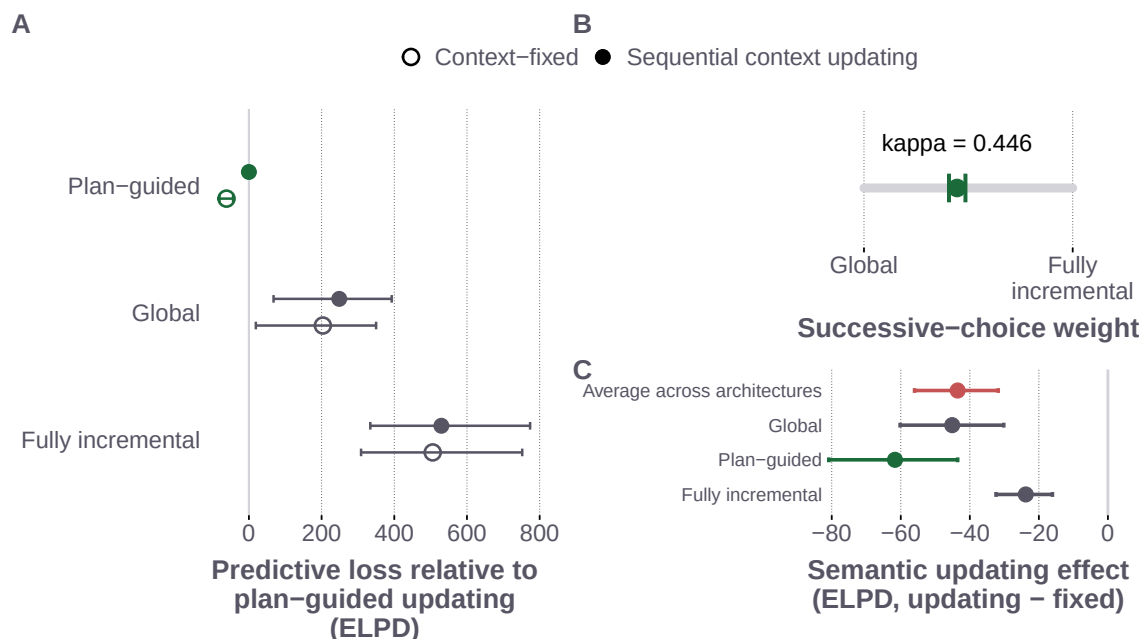


Figure 8: Matched comparisons of production architecture and sequential context updating. Panel A reports predictive loss for all six matched models relative to the plan-guided model with sequential context updating; lower values indicate better leave-one-out prediction. Panel B places the shared successive-choice weight κ between the global and fully incremental endpoints. Panel C shows updating-minus-fixed ELPD for each production architecture and their average. Intervals are 95% credible intervals from the participant-level Bayesian bootstrap; the vertical line in Panel C marks zero.

were greater than .999. The context-fixed model had the highest ELPD. The sequentially updating version was lower by 54.140 ELPD, with a 95% credible interval for updating minus fixed of $[-69.042, -39.733]$. Subsequent best-model diagnostics use the context-fixed model.

The population location of its participant-specific successive-choice weights was intermediate between the global and fully incremental endpoints, $\mu_\kappa = .472$, with a 95% credible interval of $[.337, .618]$. The participant distribution was broad, with an estimated logit-scale standard deviation of 2.720, 95% credible interval $[2.277, 3.207]$. The model had an absolute ELPD of $-13,039.924$, zero divergences, maximum $\hat{R} = 1.0051$, no observation with Pareto $k > .7$, and minimum BFMI of .742. It met the specified numerical thresholds, although 77.7% of transitions reached the depth-6 step cap, which remains a sampling-efficiency limitation.

To locate the semantic contrast, we factor the leave-one-response-out predictive distribution into response length, adjective set conditional on length, and order conditional on the adjective set. All three factors use the same posterior weights obtained by leaving out the complete response, so their log scores sum to the original response score. For the participant-weight plan-guided pair, updating minus fixed was -21.549 ELPD for length, 95% credible interval $[-30.933, -12.443]$, and -31.068 for adjective set conditional on length, $[-39.202, -23.530]$. The conditional-order contribution was -1.524 ELPD, with an interval of $[-4.879, 1.774]$. The predictive disadvantage of updating is therefore concentrated in length and adjective content; the aggregate conditional-order contrast remains uncertain. This decomposition and its counterparts for the matched architectures appear in Figure 13.

Figure 9 evaluates its adequacy for the complete observed distribution. Panel A compares frequent completed utterances in the size-colour trials, where the central ordering contrast arises. Panel B extends the check to all fifteen completed utterances across the three adjective families, referential contexts, and size-discriminability conditions. Across these 270 cells, the posterior predictions correlated with the observed proportions at $r = .909$, with a mean absolute error of 3.0 percentage points and a root mean squared error of 6.0 percentage points. The corresponding 95% posterior-predictive intervals contained 58.5% of the observed cell proportions, a descriptive check of fit across the dependent response cells. Panel C shows that the model captures the broad shifts in size-initial responses and redundant adjective use across referential contexts. A remaining discrepancy concerns high-discriminability size-form displays where both properties are necessary: size-initial responses occurred on 59.5% of trials, compared with a prediction of 45.2%, while form-initial responses occurred on 4.2% of trials, compared with 13.5% predicted. These errors locate the remaining discrepancy in initial-adjective choice.

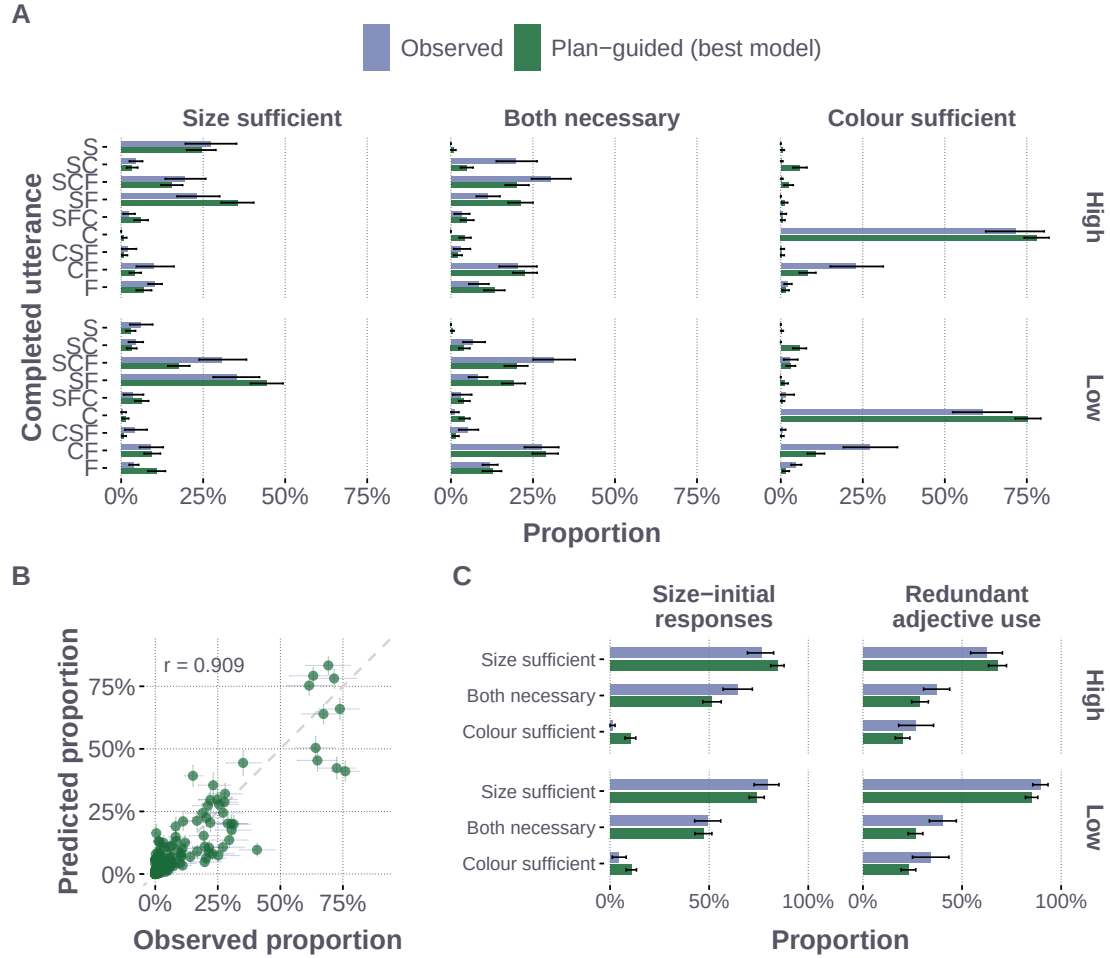


Figure 9: Posterior-predictive adequacy of the best-fitted plan-guided model. Panel A shows observed and posterior-predicted proportions with their respective uncertainty intervals for frequent size-colour utterances. Panel B compares all 270 cells; horizontal intervals show participant-bootstrap uncertainty in the observations, vertical intervals show 95% posterior-predictive uncertainty, and the dashed line marks equality. Panel C compares observed and posterior-predicted size-initial responses and redundant adjective use, with referential context on the vertical axis and size discriminability in facet rows. Intervals in Panel C show participant-bootstrap 95% intervals for the observed proportions and 95% posterior-predictive intervals for the model predictions.

5.4 Plan-Guided Production and Participant Variation

Architecture-level response patterns. Qualitative diagnostics localise the distributional patterns that distinguish plan-guided production from its matched endpoints. Figure 10 shows initial-adjective cells in which the largest difference among architecture predictions is at least 6.7 percentage points and traces these differences into completed utterances. The largest separations occur in form-involving trials, where colour and form compete for the initial position. Colour-initial responses contribute 328.912 of the matched plan-guided model’s 402.677-ELPD advantage over the average endpoint, or 81.7% of the summed pointwise contrast. Across the displayed cells, plan-guided production often lies between the global and fully incremental endpoints.

Successive normalisation produces this pattern in the model. The global speaker evaluates the completed utterance distribution. The fully incremental speaker also normalises among the adjectives available after each emerging prefix. Different opening adjectives create different sets of plausible continuations, so competition at later positions redistributes probability among the completed paths that share each initial adjective. This redistribution has larger consequences when several properties remain viable, especially when colour and form compete in form-involving displays. In the matched comparison, the plan-guided speaker gives this prefix-specific competition a shared intermediate weight. The participant refinement estimates individual weights.

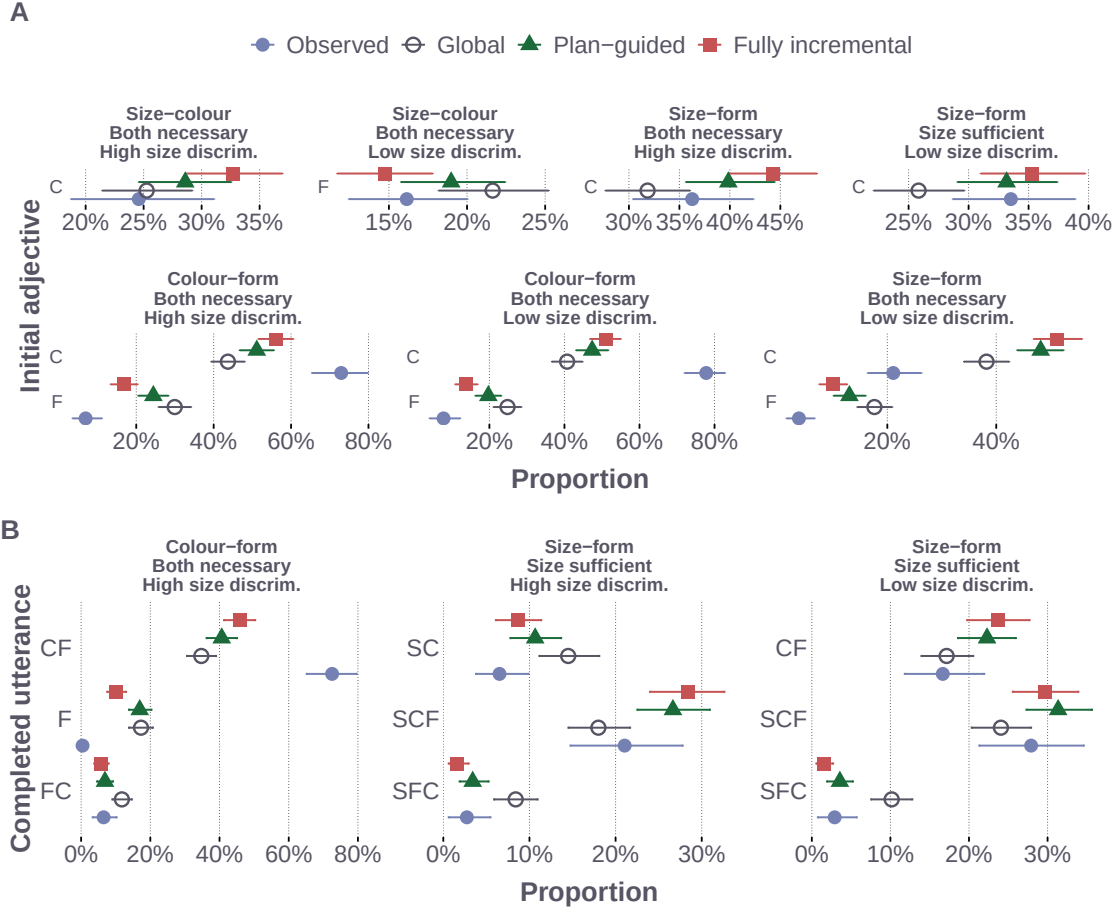


Figure 10: Qualitative comparison of the matched production architectures. Panel A shows initial-adjective cells for which the maximum difference among model predictions is at least 6.7 percentage points. Panel B shows completed utterances with differences of at least five percentage points within the three experimental conditions with the largest pairwise total-variation distance among model predictions. Response labels use S for size, C for colour, and F for form. Intervals show participant-bootstrap 95% intervals for observed proportions and 95% posterior-predictive intervals for model predictions. Horizontal scales vary across conditions to make the comparatively small differences among production architectures visible. The displayed cells are descriptive diagnostics; the held-out comparison in Figure 8 provides the evidence for relative predictive accuracy.

Participant variation. The gain from participant-specific successive-choice weights also has a clear distributional location. Participants whose posterior mean κ_i lay farther from the shared $\kappa = .446$ gained more participant-level ELPD from the hierarchy. The observed correlation was $r = .618$, and the participant-level Bayesian-bootstrap mean was .639, with a 95% credible interval of [.525, .763] and posterior probability above zero greater than .999. At the response level, the hierarchy gained 229.372 ELPD on two-adjective utterances, 95% credible interval [163.012, 319.334], and 183.796 ELPD on three-adjective utterances, 95% credible interval [100.757, 304.826]. Its contribution to one-adjective utterances was -48.426 ELPD, with a 95% credible interval of $[-70.371, -28.666]$. The participant hierarchy therefore improves prediction where speakers traverse multi-adjective paths and successive alternatives can redistribute probability among completed utterances.

The joint participant hierarchy includes variation in pragmatic optimality α_i and stable-order weight β_i alongside variation in κ_i . Posterior-predictive contrasts between participants in the lower and upper tails of each parameter localise their fitted response signatures. Higher α_i concentrates probability on a sufficient single adjective, especially S in size-sufficient displays and F in form-sufficient displays. Higher κ_i mainly redistributes probability among form-involving responses whose continuation sets differ after the first adjective. Higher β_i shifts probability from single-adjective responses towards longer responses that follow the stable order score. The posterior participant-level association between κ_i and β_i was positive, with mean $r = .162$ and 95% credible interval [.073, .249]. The associations of α_i with κ_i and β_i were centred near zero. These tail contrasts describe how the three parameters organise response variation within the joint fit.

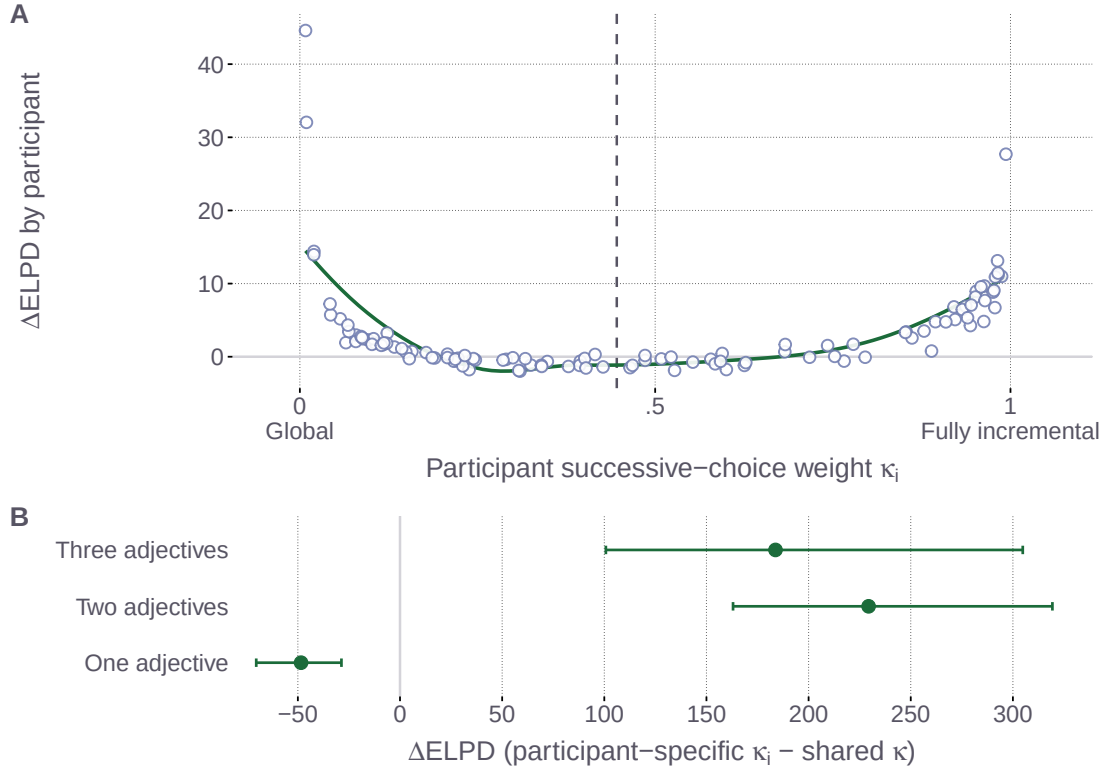


Figure 11: Where participant-specific successive-choice weights improve prediction. Panel A relates each participant’s leave-one-out gain from the κ_i hierarchy to their posterior mean κ_i ; the dashed line marks the shared- κ estimate and the curve is a descriptive smooth. Panel B decomposes the total ELPD gain by observed response length. Intervals in Panel B are 95% credible intervals from the participant-level Bayesian bootstrap. Both panels use the best-fitted plan-guided model and its otherwise matched shared- κ ablation to localise the held-out model contrast.

Maximal three-adjective utterances accounted for 1,633 of the 9,100 modelled responses, or 17.9%. Within the fitted model, these responses balance accumulated referential value and salience against a shared three-adjective cost, with α_i , κ_i , and β_i determining how that balance is expressed for each participant. The predictive gain on three-adjective utterances shows that participant variation in successive adjective choice helps locate these responses in the observed distribution. Together, the architecture and hierarchy comparisons explain how successive competition redistributes probability among complete responses and how its contribution varies across participants. The response partitions locate this gain by summing exact-response log scores within the observed categories. The shared task specification makes these contrasts comparable across models, while the semantic factorisation above separates the predictive contributions of length, adjective set, and conditional order.

6 Simulation and Generative Conditions for an Order Asymmetry

The model comparisons in Section 5 found that context-fixed semantics predicted the observed production distribution more accurately than sequential context updating within hierarchical semantic composition. The predictive disadvantage of updating was concentrated in adjective content and length, while its aggregate conditional-order contribution remained uncertain. This leaves open which computational conditions generate an order asymmetry in listener target recovery and whether that asymmetry distinguishes the semantic regimes.

That possibility is motivated by Franke, Gregory Scontras, and Simoncic (2019), whose large-scale Monte Carlo simulations found an average referential advantage for adjective orders that place more subjective adjectives farther from the noun. Their simulation averaged the referential success of two fictitious communities committed to opposite orders across randomly generated contexts. Its explanatory target was a communicative pressure that could accumulate over the longer timescale of language evolution and gradually stabilise ordering preferences. Speakers’ online deliberation about the locally optimal order and RSA-style reasoning over alternative utterances in a particular reference game lay outside that scope.

The present simulation brings their comparison across randomly generated contexts into a local decision problem, using a pragmatic listener to evaluate the two size–colour orders. The central question is which computational conditions make the average size-first advantage depend on semantic updating. The comparisons preserve the same random scenes and target-recovery measure while changing complete-utterance versus prefix evaluation, the utterance space, and semantic parameters. First, we define the simulation procedure and the measure of communicative success. Then, we identify which of these conditions preserve, weaken, or reverse the order asymmetry.

6.1 Simulation Design and Size-First Advantage

The main comparison isolates semantic composition, prefix evaluation, and utterance competition from stable-order expectations, visual salience, task-specific response patterns, and participant variation. These components are held absent while the semantic regimes are compared at common parameter settings. Additional controls for local choice and a stable-order score appear in Appendix H.1.

The central manipulation varies the number and size distribution of the objects in randomly generated displays. Each display contains between 2 and 30 objects whose sizes are drawn from a normal distribution centred at 15.5 and truncated to the interval $[1, 30]$. The largest object is designated as the target and assigned the colour blue, ensuring that both size and colour apply. Each non-target object is independently assigned blue or non-blue with equal probability, so the number of blue distractors varies with context size and across randomly generated displays. Across the generated contexts, colour more generally restricts the comparison class to a subset whose size varies from display to display. This extends the analysis beyond the experiment’s controlled sufficiency conditions and tests how the strength of a colour-based restriction affects the subsequent interpretation of size. Three levels of contextual size spread generate displays whose objects range from closely clustered to widely separated in size. This provides a simulation analogue of the experimental size-discriminability manipulation: greater spread tends to separate the largest target more clearly from its size competitors.

The simulation additionally varies colour reliability, size-threshold selectivity, and perceptual blur. These respectively control the semantic strength of the colour-based restriction, how selectively *big* applies, and how sharply size differences are represented. The same 24,000 random displays are reused across the semantic regimes, evaluation rules, utterance spaces, and parameter settings. Pragmatic optimality is fixed at $\alpha = 1$, the main comparison has no order or length costs, and the complete parameter grid and sample counts appear in Appendix H.

Both surface orders are interpreted through hierarchical semantic composition, with the noun-adjacent adjective applied first. Under context-fixed semantics, the size threshold is computed from the full scene and remains fixed throughout composition. Under sequential context updating, the threshold is recomputed from the context left by the inner adjective. For *big blue*, colour first restricts the context to blue objects before size is interpreted; for *blue big*, size is interpreted relative to the full scene before colour applies.

The initial comparison uses the two completed orders, *big blue* and *blue big*. A terminal-evaluation model scores each completed expression by the literal listener’s target probability. A prefix-evaluation model accumulates this value over the surface prefixes of the expression:

$$U_{\text{term}}(u, o, X) = \alpha \log L(o \mid u, X), \quad (11)$$

$$U_{\text{prefix}}(u, o, X) = \alpha \sum_{t=1}^{|u|} \log L(o \mid u_{1:t}, X). \quad (12)$$

Both models globally normalise their scores over the available completed responses. The prefix model corresponds to the $\kappa = 0$ restriction of the production architecture, with the additional task and order terms removed. The prefix comparison first retains the two-order utterance space, then adds the one-adjective responses *big* and *blue*, and finally expands to the fifteen responses used in production. Form semantics is fixed at .50, as in the production models. A final comparison retains the fifteen-response prefix model and replaces the parameter grid with the constants used in production: colour reliability .59, threshold selectivity .50, and perceptual blur .6856. The original two-order global and incremental simulation is retained as a reference in Appendix H; its speaker and continuation rules are specified there.

The pragmatic listener L^{prag} then applies Bayes’ rule to the speaker distribution: an object receives more probability when a speaker intending that object would be more likely to produce the observed order. All objects have equal prior probability. For each simulated context, the analysis computes this target probability after *big blue* and after *blue big*. The size-first advantage is

$$\Delta_{\text{SF}} = L^{\text{prag}}(o^* \mid \text{“big blue”}) - L^{\text{prag}}(o^* \mid \text{“blue big”}). \quad (13)$$

Positive values indicate a higher model-assigned target probability for the size-first order. The measure therefore compares pragmatic target recovery for two fixed utterances.

6.2 Conditions under which an Order Asymmetry Emerges

Figure 12 shows the controlled comparisons on the same random scenes. For terminal evaluation under context-fixed semantics, the two orders have the same interjective meaning and equal cost, so they yield the same target probability. Sequential context updating can break this symmetry because *big blue* evaluates size relative to the comparison class established by colour. Across the original parameter grid, the mean size-first advantage under terminal evaluation was 1.141 percentage points with updating and zero with fixed semantics.

Prefix evaluation changes the symmetry condition. Even when the completed expressions have identical fixed meanings, their first adjectives can have different target probabilities, which enter the accumulated prefix scores. With only the two completed orders available, the mean advantage became negative under both semantic regimes: -2.718 percentage points with fixed semantics and -1.838 with updating. Adding the one-adjective responses changed the mean direction to size-first, with advantages of 2.257 and 2.772 percentage points, respectively. Expanding to fifteen responses retained positive mean advantages of 2.589 and 3.104 percentage points. The other available expressions affect the speaker’s response normalisation for each referent, which enters the pragmatic listener’s inference about the target. Consequently, the direction of the ordering advantage depends on the utterance space even when the scenes and adjective meanings remain unchanged.

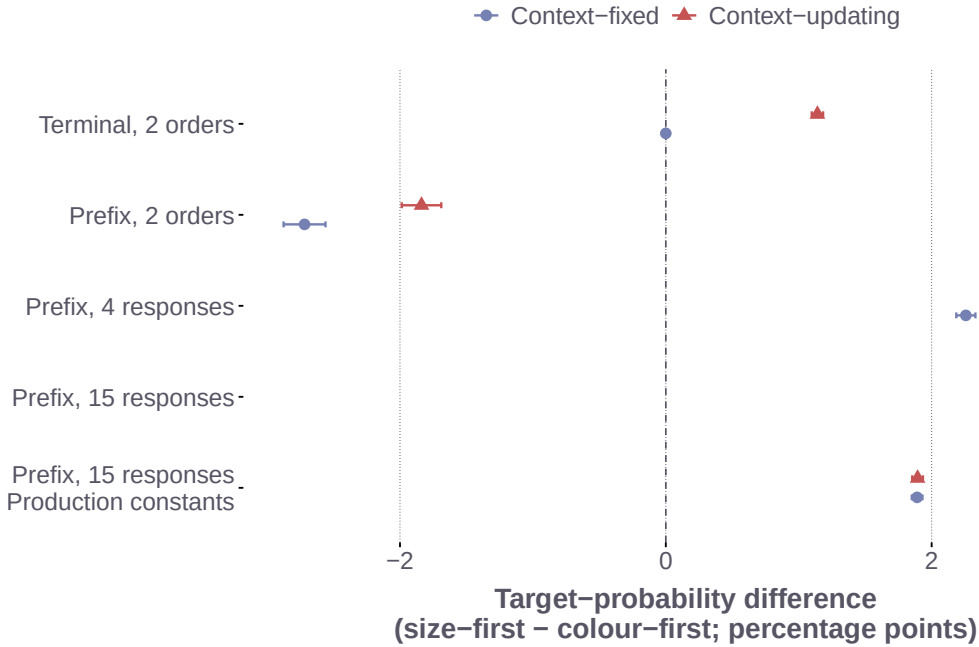


Figure 12: Controlled changes in the size-first pragmatic target-recovery advantage on the same 24,000 random scenes. The first four rows use the original semantic parameter grid; the final row uses the constants fixed in the production models. Terminal evaluation scores the completed expression, whereas prefix evaluation sums the scores of its surface prefixes ($\kappa = 0$). The two-order space contains size-colour and colour-size; the four-response space adds the two single adjectives; the fifteen-response space is the production utterance space. All rows use the production semantic implementation, $\alpha = 1$, a uniform referent prior, and no stable-order, salience, task-policy, or lapse terms. Points show scene-averaged target-probability differences in percentage points; intervals are ± 1.96 Monte Carlo standard errors, with parameter settings averaged within each scene before estimating uncertainty. Positive values favour size-first. These intervals quantify scene-sampling uncertainty, conditional on the specified models and parameter settings.

Using the semantic constants fixed in the production models brought the two fifteen-response prefix models close together: the mean advantages were 1.891 percentage points with fixed semantics and 1.894 with updating. The paired updating-minus-fixed difference was 0.0037 percentage points, with a Monte Carlo interval of $[0.0007, 0.0067]$. The full contrasts across local-normalisation weights, semantic settings, and the added stable-order score are reported in Appendix H.1. The original sweep and its scene-dependent reversals remain in Appendix H.

These comparisons identify conditions under which semantic updating generates an ordering advantage and show that the same direction can arise under fixed semantics when prefix evaluation and a broader utterance space

are available. A size-first advantage by itself therefore does not uniquely identify semantic updating. The simulation constrains the computational interpretation of an ordering asymmetry; the response-score decomposition in Appendix C.1 separately locates the empirical predictive disadvantage of updating in adjective content and length. A supplementary calculation using the fitted models on the experimental displays is reported in Appendix H.2.

7 General Discussion

The behavioural studies place adjective ordering within the broader problem of reference generation. Explicit slider judgements exhibit a replicated context gradient: size-first preference is strongest when size identifies the target, weakest when colour does, and remains slightly positive in colour-sufficient scenes. Production distributes the same contextual pressure across a larger response space. Speakers often select one sufficient adjective, and visual size reliability primarily affects whether they add material. On those trials, selection and utterance length determine whether a multi-adjective ordering decision arises at all.

The final production comparison evaluates that complete response space. With participant-specific variation in pragmatic optimality and stable order expectations controlled, plan-guided incremental production outpredicts both global and fully incremental production. The population location of the successive-choice weights lies between these endpoints, and participants vary widely around it. The matched architecture advantage is concentrated in colour-initial responses, and the participant hierarchy gains predictive accuracy primarily on multi-adjective utterances. Thus, the architecture result and the hierarchy result converge on successive adjective competition as a useful component of complete-response prediction.

Stable order expectations enter through a German language-model residual whose influence varies across participants. The model treats this conventional size-first tendency as a controlled source of variation. The model additionally includes salience and a task link for adjective sufficiency, form use, visual reliability, and utterance length. These components connect the observed task choices to the common response space used by the architecture comparison. Plan-guided production operates within a broader account of the task decisions that make an adjective order observable.

The two directional accounts of communicative efficiency receive different forms of evidence. Context-fixed semantics predicts the complete response distribution more accurately than sequential context updating within hierarchical semantic composition in the present materials. The diagnostic decomposition locates this difference primarily in length and adjective content, with an uncertain aggregate conditional-order contribution. Controlled simulation shows that updating can generate a size-first target-recovery advantage under terminal evaluation. The same direction also arises under fixed semantics when prefix evaluation is combined with a broader utterance space, and the two regimes yield close mean advantages at the semantic constants used in production. The simulation therefore identifies conditions for generating the asymmetry and limits its ability to distinguish semantic updating from other computational sources. Together, the fitted comparison and simulation show why joint modelling of selection, length, and order is central to evaluating the two proposed directions of communicative efficiency.

7.1 Simulation Conditions and Online Processing

The simulation makes the interpretation of a size-first advantage conditional on the speaker's evaluation rule, the utterance space, and semantic parameters. Under terminal evaluation, updating must meaningfully reshape the comparison class for size to produce an asymmetry. Under prefix evaluation, the opening adjective can contribute an asymmetry even with fixed meanings, and adding single-adjective alternatives can reverse its average direction. These results connect the simulation to the production study's central design choice: adjective selection, utterance length, and order need to be evaluated within a common response space. The presence of a size-first advantage alone provides limited leverage for identifying semantic updating, and the empirical predictive disadvantage of updating concerns mainly other parts of the response distribution.

Treating the semantic-updating route as an online production mechanism carries stronger processing commitments. For a speaker to exploit the semantic advantage of *big blue*, an advance utterance plan must make the later, noun-adjacent colour adjective available before size is produced. The speaker would then need to filter the referential context to colour-matching objects, recompute the size standard within that restricted set, and evaluate the resulting communicative value against alternative utterances. This interpretation links the semantic computation to the availability of later adjectives during planning. The computation may be approximated during online planning, and its average communicative advantage may accumulate across repeated language use into stable ordering expectations, as envisaged by Franke, Gregory Scontras, and Simonin (2019). Planning-time manipulations and online measures such as speech onset and eye movements would be required to determine whether speakers perform the restriction and comparison during production (Ferreira and Swets, 2002; Konopka and Meyer, 2014).

The context-fixed fully incremental speaker can produce the same size-first surface order through local informativity. When size is perceptually clear, it is informative at the first adjective position and can be selected without recomputing a colour-restricted comparison class. An egocentric incremental speaker could yield the same qualitative pattern by selecting a visually accessible property early in planning (Pechmann, 1989; Engelhardt, K. G. Bailey, and Ferreira, 2006). Related work shows how early informativity can shape incremental reference production and word order (Rubio-Fernandez and Jara-Ettinger, 2020; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021). This route connects early adjective choice to the visual information available at the start of production. In the original two-order simulation, this local pressure produces colour-first reversals when size becomes less discriminating than colour. The controlled prefix comparisons also produce such reversals, showing that their occurrence depends on the evaluation rule and available responses. The production comparison favours combining these locally informative choices with complete-utterance evaluation: plan-guided production outpredicts the fully incremental endpoint. The same surface order can therefore arise from hierarchical context updating, locally incremental choice, or stable order expectations, and completed utterances alone do not identify which computation a speaker performs online.

7.2 Participant Variation and Redundant Production

The participant hierarchy locates individual variation along pragmatic optimality, successive adjective choice, and stable order expectations. Maximal three-adjective descriptions make this variation particularly visible because they extend beyond the material sufficient for identification. Egocentric accounts predict redundant production when perceptually accessible properties are encoded early and omitting them would require additional planning (Pechmann, 1989; Engelhardt, K. G. Bailey, and Ferreira, 2006). Audience-oriented accounts predict redundant production when additional properties can guide visual search or hedge uncertainty about the cues available to the listener (Rubio-Fernández, 2016; Rubio-Fernandez, 2019; Degen et al., 2020; Tourtouri, Delogu, and Crocker, 2021). Both pressures can converge on the same completed utterance in a shared visual task. Independent manipulations of property accessibility, partner knowledge, and listener uncertainty would be required to distinguish their contributions. The participant parameters and response signatures in Section 5 provide distributional targets for these tests.

7.3 Inference and Computational Constraints

Participant-specific weights earn the largest predictive gains for speakers whose posterior mean κ_i lies farthest from the shared estimate (Figure 11A). The broad logit-normal hierarchy maps substantial latent variation towards the endpoints of the κ scale. This continuous distribution accommodates the observed heterogeneity without requiring discrete groups of global and fully incremental speakers. Repeated observations across tasks would clarify how stable these response profiles are for individual speakers.

The production model also exposes a computational challenge for extending this account to larger vocabularies (Degen, 2023). With d available adjectives and no repetition, the number of ordered non-empty strings is $|\mathcal{U}_d| = \sum_{k=1}^d d!/(d-k)!$: three adjectives yield 15 responses, five yield 325, and ten yield almost ten million. Every response is evaluated across its prefixes and possible referents. Precomputation, vectorised JAX operations, and GPU inference make this calculation feasible for the present task; larger spaces motivate approximations to utterance retrieval and local competition (Rooij, 2008; Kwisthout, Wareham, and Van Rooij, 2011). Sampling salient utterances or retaining a limited set of continuations would provide concrete ways to relate plan-guided production to resource-rational computation (Lieder and Griffiths, 2020).

The resulting question is how much predictive accuracy can be retained as the computational cost decreases. This connects the trade-offs among referential success, early informativeness, stable order expectations, and utterance length to accounts of pragmatic reasoning as a balance between utility and effort (Zaslavsky, Hu, and Levy, 2021). Comparing approximate models against the present exact calculation would make that trade-off measurable.

7.4 Generalisation Beyond the Present Task

The experiments use German prenominal noun phrases in a controlled reference task with a fixed adjective vocabulary and a restricted fifteen-response support. The form and task specifications and the fixed GPT-2 order feature provide the common setting for the reported architecture comparisons. Model development used these data, and response-wise LOO evaluates the resulting selected specifications. The present evidence therefore motivates a further test on new participants and materials. Further work can test whether the architecture generalises to freer production, new lexical families, different adjective systems, and designs that independently manipulate lexical

subjectivity and perceptual reliability. Planning-time manipulations would complement those predictive tests by examining how complete-utterance evaluation and successive choices are implemented during production.

Taken together, the results locate adjective order within choices about what to mention and how much to say. Plan-guided incremental production provides the best predictions by combining complete-utterance evaluation with competition among successive adjective choices. The resulting account connects stable ordering preferences to the context-sensitive decisions that determine which adjectives are produced in the first place.

8 Data Availability Statement

Data, analysis scripts, model code, final posterior summaries, and figure-generation scripts will be made available in an anonymised repository upon submission. The repository will include the deterministic production input, the final global, plan-guided, and fully incremental model artifacts, preprocessing and reconstruction receipts, and the scripts required to reproduce the reported behavioural and predictive analyses.

9 Author Note

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A Methods and Coding Transparency

Table 1 summarises the participant exclusions and the analysis tables used in the reported manuscript analyses. The per-criterion exclusion counts are not mutually exclusive, so the unique number of excluded participants is smaller than the sum of the criterion counts. For Study 2, the canonical production table contains all three adjective-pair families. The production model comparison reported in Section 5 uses all 9,100 canonical annotated observations over the 15 utterance categories and three adjective-family pairings. The size–colour marginal is retained as the theoretical anchor and descriptive subset.

Table 1: Participant exclusions and analysis-table sizes. Criterion counts overlap within study.

Experiment	Starting and final N	Participant exclusions	Analysis table used in manuscript
Study 1a slider ratings	150 → 134	8 with more than three missing trial-list entries; 8 with extreme completion time; 2 with more than three control-item failures; 6 with more than three “none fits” responses on experimental items; 16 unique participants excluded.	Broad cleaned critical table: 10,606 observations. Reported size–colour slider subset: 3,485 observations.
Study 1b slider replication	132 → 104	6 missing-list failures; 1 duration failure; 3 control failures; 24 experimental none-fit failures; 28 unique participants excluded.	Cleaned replication table: 8,379 critical observations. Reported size–colour slider subset: 2,786 observations.
Production study	149 → 113	8 with extreme completion time; 25 with more than 80% identical responses; 7 with more than 10% missing or unannotatable responses; 36 unique participants excluded.	Canonical full-data table: 9,100 annotated observations. Reported size–colour behavioural subset: 3,034 observations.

The stimulus inventory contains six lists of 180 main trials: 81 critical trials and 99 fillers and controls per list, plus three practice displays. Lists 1–3 use high size discriminability and lists 4–6 use low size discriminability. Each list contains nine critical trials for each combination of the three adjective families and three referential conditions. The stored large-size band is {9, 10} in both groups; the smaller-size band is {1, 2, 3} at high discriminability and {1, 2, 3, 4, 5, 6} at low discriminability. Thus, contrasts between the large and smaller bands can span 6–9 recorded units in the high-discriminability materials and 3–9 units in the low-discriminability materials. The released stimulus table specifies the six object sizes, colours, and forms for every list–item–condition combination.

The slider-rating materials used the fixed adjective vocabulary in Table 2. The raw stimulus table encodes colours and forms with compact feature labels; the interface displayed the corresponding German adjective forms in phrases such as *den großen blauen Aufkleber*.

Table 2: Adjective vocabulary in the slider-rating item descriptions.

Property class	Raw stimulus values	German adjective forms displayed in the item text
Size	Numeric size levels 1–6, 9, 10	<i>großen, kleinen</i>
Colour	blue, orange, grey, brown, green, black	<i>blauen, orangen, grauen, braunen, grünen, schwarzen</i>
Form	circle, diamond, heart, quadrat, star, triangle	<i>runden, diamantförmigen, herzförmigen, quadratischen, sternförmigen, dreieckigen</i>

The final modelling table is reconstructed deterministically from the raw stimulus records. Observed colour and form values are recoded into target–object relations without stochastic nuisance-feature generation; this recoding is separate from the lexical forms shown to participants.

The first author coded production responses into the 15 ordered adjective categories listed in Table 3. The data files use D for the size or dimension adjective. The model exposition uses S; the two notations refer to the

same adjective class. Misspellings and transparent lexical alternatives were normalised when their semantic class was unambiguous. Spatial descriptions, comparative or ordinal size expressions, and responses that could not be assigned to a target adjective class were excluded from the production model comparison.

Table 3: Production coding dictionary. In the data files, D corresponds to the size adjective S; C and F correspond to colour and form.

Length	Codes
One adjective	D, C, F
Two adjectives	DC, DF, CD, CF, FD, FC
Three adjectives	DCF, DFC, CDF, CFD, FDC, FCD

The behavioural hypotheses were evaluated with Bayesian hierarchical models. All analyses used the size–colour subsets reported in the main experiment figures: 3,485 observations in Study 1a, 2,786 in Study 1b, and 3,034 in Study 2. The later production-model comparison addresses predictive differences among the proposed computational mechanisms over the complete fifteen-category response space and all 9,100 production trials. For the slider analyses, ratings were reoriented toward size-first, divided by 100, and centred by subtracting 0.5. Let $\mathbf{x}_{ij}$ encode referential context, size discriminability, and their interaction, and let $\mathbf{z}_{ij}$ encode the participant intercept and context slopes. The slider models used a Gaussian likelihood with

$$y_{ij} \sim \text{Normal}(\mu_{ij}, \sigma), \quad \mu_{ij} = \mathbf{x}_{ij}^T \boldsymbol{\beta} + \mathbf{z}_{ij}^T \mathbf{b}_i, \quad \mathbf{b}_i \sim \text{MVN}(\mathbf{0}, \boldsymbol{\Sigma}_{\text{participant}}).$$

For production, size-initiality indicated whether the coded response began with the size adjective. Over-informativeness indicated a three-adjective response in both-necessary contexts or a multi-adjective response in one-property-sufficient contexts. Both binary outcomes used a Bernoulli likelihood with a logit link and an item intercept c_k :

$$y_{ijk} \sim \text{Bernoulli}(\pi_{ijk}), \quad \text{logit}(\pi_{ijk}) = \mathbf{x}_{ijk}^T \boldsymbol{\beta} + \mathbf{z}_{ijk}^T \mathbf{b}_i + c_k, \quad c_k \sim \text{Normal}(0, \sigma_{\text{item}}).$$

For the slider studies, the interaction was represented by two posterior contrasts: the high-minus-low discriminability effect in the both-necessary context minus the corresponding effect in the size-sufficient context, and the high-minus-low effect in the colour-sufficient context minus the effect in the size-sufficient context. The production interactions were planned contrasts on the posterior-predictive probability scale. For size-initiality, the contrast was the high-minus-low discriminability effect in the both-necessary context minus the mean effect in the two one-property-sufficient contexts. For over-informativeness, the contrast was the low-minus-high discriminability effect in the size-sufficient context minus the mean effect in the both-necessary and colour-sufficient contexts.

The slider models used a Gaussian likelihood, correlated participant intercepts and context slopes, fixed-effect priors $\text{Normal}(0, .30)$, $\text{HalfNormal}(.20)$ priors on participant standard deviations, and a $\text{HalfNormal}(.25)$ prior on residual standard deviation. The production models used Bernoulli likelihoods, the same correlated participant effects, an item intercept, a $\text{Normal}(0, 2)$ intercept prior, $\text{Normal}(0, 1.5)$ priors on the remaining fixed effects, and $\text{HalfNormal}(1)$ priors on participant and item standard deviations. Participant-effect correlations received an LKJ(2) prior. Four vectorized chains each provided 1,000 retained draws after 1,000 warm-up iterations. All four fits had zero divergences, no maximum-tree-depth hits, $\hat{R} \leq 1.01$, and minimum effective sample sizes above 900. For each posterior draw, posterior-predictive cell expectations were computed for a new participant and, in production, a new item by integrating over the fitted random-effects distributions. The planned predictive hypotheses were then evaluated on the rating or response-probability scale. The analysis records cell predictions, planned hypothesis contrasts, and inference diagnostics for all four fits.

B Final Production Model Specification

B.1 Deterministic Input Reconstruction

The final production input is reconstructed one-to-one from the raw stimulus table using participant list, condition, and item identifiers. Each retained response is paired with one observed 6×3 display tensor containing size, colour, and form values. An earlier preprocessing pipeline generated nuisance-feature values stochastically for some nominally irrelevant dimensions; the final analysis removes this transformation. Legacy first-draw, last-draw, and agreement-subset analyses are retained as historical preprocessing checks.

B.2 Graded Size Meaning

Size is represented by a graded threshold over the scene-level comparison class $\mathcal{C}$ (Schmidt et al., 2009; Franke, Gregory Scontras, and Simoncic, 2019):

$$\theta_\delta(\mathcal{C}) = Q_{.80}(\mathcal{C}) - \delta (Q_{.80}(\mathcal{C}) - Q_{.20}(\mathcal{C})), \quad (14)$$

where $Q_{.20}$ and $Q_{.80}$ are probability-weighted size quantiles and δ controls threshold selectivity. For an object of size s_o , graded membership is

$$m_S(o | \mathcal{C}) = \Phi \left(\frac{s_o - \theta_\delta(\mathcal{C})}{w \sqrt{s_o^2 + \theta_\delta(\mathcal{C})^2 + \epsilon}} \right), \quad w = .6856, \quad \epsilon = 10^{-8}, \quad (15)$$

where Φ is the standard normal cumulative distribution function. The weighted quantiles are the first observed sizes at which cumulative comparison-class probability reaches .20 and .80. The same width w is used in both discriminability groups; the recorded sizes enter this function directly. For colour and form, matching objects receive membership ν and nonmatching objects $1 - \nu$, with $\nu_C = .59$ and $\nu_F = .50$. Each prefix is interpreted from the uniform object prior, applying its modifiers once in noun-outward order. Context updating changes the comparison-class weights used for the size quantiles as inner modifiers are applied. Equations (1), (2), (4) and (6) define how this meaning enters the literal listener L and the three production accounts.

B.3 Component Inventory

Table 4: Final production-model components.

Component	Status	Role
Observed display input	Fixed	All visible size, colour, and form values
Size threshold and blur	Fixed constants	Graded size meaning in the scene comparison class
Colour reliability	Fixed	Soft determinate colour meaning
Form reliability	Fixed neutral	Form production supported by the task link
Pragmatic rationality	Estimated, hierarchical	Participant variation in response concentration
Stable-order score Ω	Fixed feature	Relative order expectation within adjective sets
Stable-order weight β_i	Estimated, hierarchical	Participant variation in stable order expectation
Salience coefficient	Estimated	Display prominence in candidate utility
Task-link coefficients	Estimated	One-word support, form backup, length and reliability terms
Successive-choice weight κ_i	Estimated, hierarchical, in plan-guided production	Participant variation in successive adjective choice

The population pragmatic-optimality parameter has prior $\alpha_0 \sim \text{HalfNormal}(3)$, and the population log stable-order weight has prior $\mu_{\log \beta} \sim \text{Normal}(0, .35)$. Participant effects are noncentred: $\alpha_i = \alpha_0 \exp(\tau_{\log \alpha} z_{\alpha i})$ and $\beta_i = \exp(\mu_{\log \beta} + \tau_\beta z_{\beta i})$, with independent standard-normal z values, $\tau_{\log \alpha} \sim \text{HalfNormal}(.15)$, and $\tau_\beta \sim \text{HalfNormal}(.5)$. The shared plan-guided weight has prior $\kappa \sim \text{Uniform}(0, 1)$. In the participant-weight refinement, $\mu_\kappa \sim \text{Uniform}(0, 1)$ and $\kappa_i = \text{logit}^{-1}(\text{logit}(\mu_\kappa) + \tau_\kappa z_{\kappa i})$, with $\tau_\kappa \sim \text{HalfNormal}(.5)$ and independent $z_{\kappa i} \sim \text{Normal}(0, 1)$. The salience coefficient has a $\text{HalfNormal}(1)$ prior and the salience-continuation coefficient a $\text{HalfNormal}(.75)$ prior. The sufficient-single-adjective and form-use coefficients each have a $\text{HalfNormal}(1.5)$ prior; the three-adjective penalty and the two size-reliability coefficients each have a $\text{HalfNormal}(1)$ prior. The fixed semantic constants are colour reliability .59, form reliability .50, threshold selectivity .50, and perceptual blur .6856. A final uniform lapse of .003 applies over the fifteen responses.

Visual salience. The salience feature combines a fixed preference across properties with their contrast in the displayed scene. For the target o^* , let s_o , c_o , and f_o be an object’s recorded size, colour, and form, and let h indicate high size discriminability. The three contrast scores are

$$r_S = \frac{1+h}{2} \text{logistic} \left(\frac{s_{o^*} - \max_{o \neq o^*} s_o}{\text{sd}(s) + 10^{-8}} \right), \quad (16)$$

$$r_C = 1 - \frac{1}{5} \sum_{o \neq o^*} \mathbf{1}\{c_o = c_{o^*}\}, \quad r_F = 1 - \frac{1}{5} \sum_{o \neq o^*} \mathbf{1}\{f_o = f_{o^*}\}. \quad (17)$$

Here $\text{sd}(s)$ is the population standard deviation across the six displayed sizes. The fixed baseline scores are $(b_S, b_C, b_F) = (0, 1, .25)$. Centring across the three properties gives

$$z_a = b_a + r_a - \frac{1}{3} \sum_v (b_v + r_v).$$

A property receives greater salience when it contrasts with the distractors; the size contrast is additionally scaled by discriminability. The same feature enters prefix utility through $\lambda_{\text{salience}} z_a$ and the continuation cost in the task link below.

Stable order expectations. We derived the fixed order feature from `dbmdz/german-gpt2`. For each pairing of six colour and six form adjectives, the scoring script generated all fifteen non-empty orders using those adjectives and *großer*, followed by the noun *Aufkleber*, without a surrounding sentence. It evaluated the resulting 540 strings, including repeated shorter strings, using mean token surprisal in bits. We averaged this surprisal within each order category and normalised $2^{-\bar{s}(u)}$ across the fifteen categories to obtain the fixed weights $q(u)$. For $\mathcal{U}_{A(u)}$, the set of orders expressing the same adjective set as u , the model uses

$$\Omega(u) = \log q(u) - \frac{1}{|\mathcal{U}_{A(u)}|} \sum_{v \in \mathcal{U}_{A(u)}} \log q(v). \quad (18)$$

This centring preserves relative order preferences within an adjective set and gives every single-adjective response a score of zero. The scoring script, archived token-surprisal table, aggregation code, and fixed scores and residuals are supplied with the model code. Participant-specific β_i determines the influence of these stable expectations when the model normalises across the complete response space.

Task link for adjective selection and length. The task link $T(u, X)$ connects the referential design to choices about which properties to mention and how far to continue a description. Let $n = |u|$ and $A(u)$ denote the length and unordered adjective set of u . For this term, X includes the display’s condition labels: $d \in \{S, C, F, \emptyset\}$ is the property that identifies the target on its own, with $d = \emptyset$ in both-necessary conditions; c indicates a colour-sufficient condition; and $h = 1$ indicates high size discriminability ($h = 0$ for low). Using $\mathbf{1}\{\cdot\}$ for an indicator, the selected models use

$$\begin{aligned} T(u, X) = & -\rho_{\text{cont}} \sum_{t=1}^{n-1} \max(z_{w_t}, 0) + \lambda_{\text{suff}} \mathbf{1}\{d \neq \emptyset, u = (d)\} \\ & + \lambda_F (1 - c) \mathbf{1}\{F \in A(u)\} - \lambda_3 \mathbf{1}\{n = 3\} \\ & + \lambda_S h \mathbf{1}\{d = S, u = (S)\} \\ & + \lambda_{SF} (1 - 2h) \mathbf{1}\{d = S, A(u) = \{S, F\}\}. \end{aligned} \quad (19)$$

The continuation cost sums the positive salience scores of adjectives followed by another adjective, using the same z scores as the prefix utility in Equation (2). The sufficiency term favours a single adjective that identifies the target, while the form term increases the value of form-containing descriptions outside colour-sufficient conditions. With form reliability fixed at .50, form gives equal literal support to matching and nonmatching objects; this task term provides a direct contribution to form use. The three-adjective penalty offsets the value accumulated along longer descriptions. In size-sufficient conditions, the final two terms additionally favour the single size adjective at high discriminability and favour size–form pairs at low discriminability while penalising them at high discriminability. The pair term applies equally to both orders. All six coefficients are nonnegative and shared across participants within each fitted model, with the priors given above. These terms enter the completed-utterance score alongside accumulated prefix utility, local competition, and the stable-order score; normalisation over all fifteen responses turns their combined values into predictions of adjective selection, length, and order.

Numerical evaluation and response likelihood. The implementation uses 64-bit arithmetic and a floor of 10^{-8} for size memberships, listener target probabilities before taking logarithms, and local choice probabilities. Colour and form memberships receive an additive 10^{-8} offset. Writing $g(u)$ for accumulated prefix utility, $i(u)$ for the sum of floored local log-choice probabilities, and $B(u) = \beta_i \Omega(u) + T(u, X)$, the computed pre-lapse score is $(1 - \kappa_i)g(u) + \kappa_i i(u) + B(u)$. Normalising these scores yields $\tilde{S}_{\kappa_i}(u | X)$, whose endpoints obey the geometric interpolation in Equation (7). The categorical response likelihood is

$$p(y_{it} = u | X_{it}) = (1 - .003)\tilde{S}_{\kappa_i}(u | X_{it}) + \frac{.003}{15}. \quad (20)$$

The matched global and fully incremental models fix κ at zero and one, respectively; the matched plan-guided model estimates a shared κ ; the final refinement estimates κ_i . All models retain participant hierarchies for pragmatic optimality and stable-order weight, with the remaining task coefficients shared across participants.

C Production Account Comparison and Diagnostics

Table 5: Bayesian-bootstrap contrasts for the matched production architectures and the plan-guided participant hierarchy.

Contrast	Δ ELPD	95% credible interval	$P(\Delta > 0)$
Plan-guided, shared κ , minus average endpoints	402.677	[311.466, 503.098]	> .999
Fully incremental minus global	-291.596	[-684.673, -1.321]	.024
Updating minus fixed, shared- κ grid	-43.527	[-56.005, -31.770]	< .001
Participant κ_i hierarchy, context-fixed	364.742	[248.178, 520.967]	> .999
Participant κ_i hierarchy, updating	372.285	[255.597, 526.056]	> .999

All eight selected production fits used four chains and 1,000 warm-up iterations per chain. The two global models retained 4,000 draws per chain at maximum tree depth 8 after the initial fits failed the mixing criteria; the other six models retained 1,000 draws per chain at depth 6. The selected fits had zero divergences, maximum $\hat{R} < 1.01$, minimum bulk and tail effective sample sizes above 400, minimum BFMI above .64, and no observation with Pareto $k > .7$. The participant-weight plan-guided fits reached the depth-6 step cap on 77.7% of transitions under fixed semantics and 31.7% under updating; this limits sampling efficiency despite the passing numerical diagnostics. Independent semantic checks verified that each modifier is applied once from the noun outwards and that changing the semantic regime can change predictions at matched parameters. Endpoint, interpolation, and direct-versus-precomputed probability checks agreed within numerical precision, and posterior likelihood replay agreed with each stored artifact within 10^{-8} .

C.1 Decomposing the Semantic Predictive Contrast

Let $\ell(u)$ denote utterance length and $A(u)$ its unordered adjective set. For each held-out response, its predictive probability factors as

$$p_{-it}(u | X_{it}) = p_{-it}(\ell(u) | X_{it}) p_{-it}(A(u) | \ell(u), X_{it}) p_{-it}(u | A(u), X_{it}). \quad (21)$$

We first integrate the fifteen-response distribution using the PSIS weights for leaving out that entire response, then obtain the length and adjective-set probabilities by summation. Taking logarithms yields three additive contributions to the original response score. This factorisation uses one common leave-out target; it does not fit or validate three separate prediction tasks. Paired differences are summed within participant, with 95% intervals from 200,000 Dirichlet-bootstrap draws using seed 20260906. Across all four semantic pairs, adjective set conditional on length makes a negative contribution to updating minus fixed. For the participant-weight pair, the original response contrast sums to -17.548 ELPD in size-colour trials, -51.172 in size-form trials, and 14.580 in colour-form trials. These descriptive partitions locate the full-response contrast and do not identify a causal effect of updating on a particular linguistic decision.

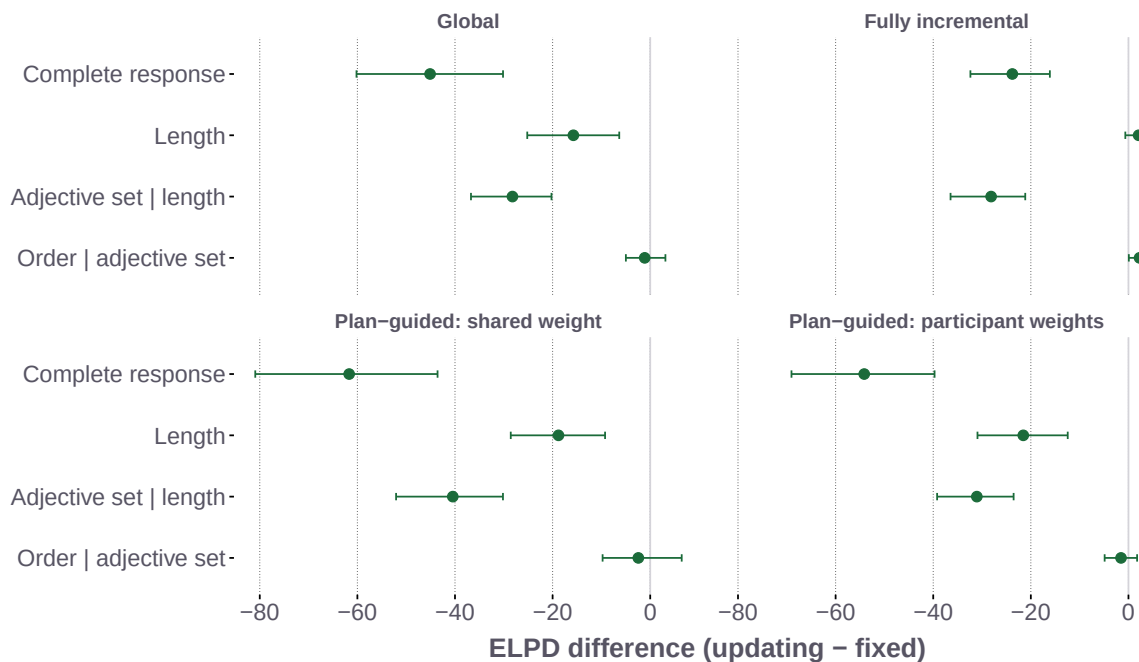


Figure 13: Additive decomposition of updating-minus-fixed response ELPD for each matched semantic pair. Length, adjective set conditional on length, and order conditional on the adjective set sum to the complete-response contrast. Points show paired score differences and intervals show participant-level Bayesian-bootstrap 95% credible intervals. The final panel uses participant-specific successive-choice weights; the other plan-guided panel uses a shared weight.

D Model-Development Checks

The final model specification was developed using the present data. Earlier exploratory comparisons informed the choice of form semantics, order feature, and task terms; the reported predictive results use the corrected primary model family with these choices held constant.

Form package. Form reliability is fixed at .50, and an estimated task coefficient supports form use. Earlier variants used informative form meanings tied to colour reliability and alternative priors on the form-use coefficient.

Stable order source. The selected specification uses the GPT-2 order residual. Earlier variants added a term for reliability-based order preferences or omitted this score.

Task link. The selected specification includes sufficient-single-adjective support, form use, a three-adjective penalty, and size-discriminability terms. An earlier reduced-policy variant removed these terms as a bundle. The auxiliary comparisons were not reconstructed with the corrected semantic implementation, so the present architecture comparison estimates performance within the selected common task specification.

E Participant Hierarchy Audit

The final joint model estimates pragmatic optimality, successive-choice weight, and stable-order weight simultaneously. Relative to the otherwise matched shared- κ model, the joint hierarchy gained 364.742 ELPD under context-fixed semantics, 95% credible interval [248.178, 520.967], and 372.285 ELPD under sequential context updating, 95% credible interval [255.597, 526.056]. Its sampler and PSIS diagnostics passed the registered thresholds. In the context-fixed joint model, the mean participant-level posterior correlations were .010 for α_i with κ_i , .025 for α_i with β_i , and .162 for κ_i with β_i . Descriptive correlations and tail-group predictions characterise the response patterns associated with the latent parameters within the fitted model.

F Endpoint Identity and Architecture Audit

At common parameter values, plan-guided production with $\kappa = 0$ exactly recovers global production, and with $\kappa = 1$ exactly recovers fully incremental production. The participant-specific order mapping, terminal response-policy kernel, and response support are identical across the three architectures. Maximum category-probability and log-probability differences were zero at a numerical tolerance of 10^{-10} .

G Sequential Context Updating for Size

For the hierarchical semantic path, let C_{j-1} be the context state after the syntactically inner modifiers have composed. The context-updating listener recomputes the size threshold from this hierarchically prior state:

$$\llbracket S \rrbracket_{\text{upd}, C_{j-1}}(o) = \llbracket S \rrbracket_{\theta_\delta(C_{j-1})}(o), \quad (22)$$

The context-fixed listener anchors the threshold to the scene prior:

$$\llbracket S \rrbracket_{\text{fix}}(o) = \llbracket S \rrbracket_{\theta_\delta(P_0)}(o). \quad (23)$$

For *big blue*, hierarchical composition applies the colour restriction before interpreting size. The distinct surface production calculation follows left-to-right order and uses the validated factorisation in which every meaning is applied once. The matched observed-display comparison favoured fixed comparison classes for predicting the complete response distribution; its decomposition appears in Appendix C.1.

H Simulation Parameter Sweeps

Figures 15 to 17 show how the three fixed semantic parameters affect communicative success across the two speaker architectures. Each figure separates the semantic regimes and averages equally over the other parameter settings. For the two-order simulation, write $q(v, o) = L(o \mid v, X)^\alpha$, with $\alpha = 1$ and equal utterance costs. For distinct adjectives a, b , the global and incremental speakers are

$$S_G(ab \mid o, X) = \frac{q(ab, o)}{q(ab, o) + q(ba, o)}, \quad (24)$$

$$R_I(ab \mid o, X) = \frac{q(a, o)}{q(a, o) + q(b, o)} \frac{q(ab, o)}{q(ab, o) + q(a, o)}, \quad (25)$$

$$S_I(ab \mid o, X) = \frac{R_I(ab \mid o, X)}{R_I(ab \mid o, X) + R_I(ba \mid o, X)}. \quad (26)$$

The second factor of R_I is the continuation-versus-stop choice, and the final normalisation conditions on producing a completed two-adjective order. The pragmatic listener uses a uniform referent prior, giving $L^{\text{prag}}(o \mid ab, X) = S(ab \mid o, X) / \sum_r S(ab \mid r, X)$. The implementation clips very small intermediate probabilities at 10^{-20} .

The reproducible sweep uses seed 20260906 and samples 1,000 independent scenes for each context-size and size-spread combination. Each scene is reused across semantic parameters, semantic regimes, and speakers, yielding 24,000 independent scenes and 12,000,000 model evaluations. Monte Carlo standard errors are calculated after averaging parameter settings within scenes; repeated evaluations of one scene do not count as independent samples. The computations use CUDA and 64-bit arithmetic.

Figure 14 shows how the size-first advantage changes across the broader random contexts, in which colour generally restricts the comparison class without uniquely identifying the target. Size spread determines how reliable size is as a cue. Widely separated sizes make the largest target stand out from its competitors and therefore make size more reliable. Closely clustered sizes make the target less distinguishable by size and therefore make size less reliable.

The simulation’s global context-fixed model provides a symmetry baseline: both orders have the same fixed intersective meaning and equal cost, so they yield the same target probability. Sequential context updating breaks this symmetry because *big blue* interprets size relative to the comparison class established by colour, producing a size-first advantage under global production. The advantage is strongest in small contexts with widely separated sizes, where restricting the comparison class has the clearest consequences for which objects count as big, and weakens as contexts grow or sizes cluster.

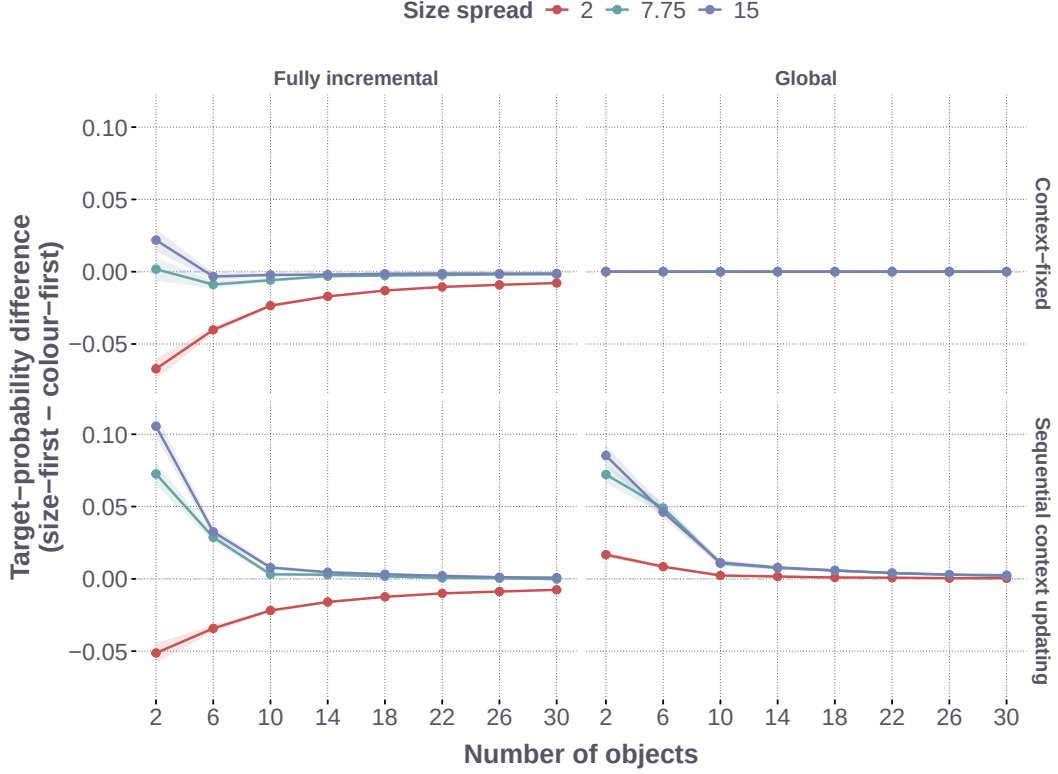


Figure 14: Size-first advantage Δ_{SF} across context size and contextual spread, shown for the global and fully incremental speakers under context-fixed semantics and sequential context updating. Wider size distributions make size a more reliable identifying cue; clustered distributions make it less reliable. Each point averages over the semantic sensitivity settings; ribbons show ± 1.96 Monte Carlo standard errors calculated after averaging those settings within each sampled scene. The complete parameter grid appears in Appendix H. Positive values indicate a higher model-assigned target probability for the size-first order. The figure shows model behaviour across the specified semantic grid.

Fully incremental production adds a pressure from local informativity along the surface sequence. When sizes are closely clustered, colour is the more informative opening cue and the incremental speaker favours colour-first. As sizes spread apart, size becomes a more reliable opening cue and the colour-first pressure weakens. After averaging over the semantic parameter settings in Figure 14, colour-first reversals occur under fully incremental production; the global context-fixed model remains symmetric, and the global context-updating model remains size-first. Individual parameter settings can yield small negative estimated advantages under global updating, so the average pattern does not establish a uniform advantage throughout the grid. Under incremental production, sequential context updating shifts the balance towards size-first when the colour-restricted comparison class changes size interpretation sufficiently strongly. The full parameter sweeps in Appendix H show that the same competition extends across colour reliability, size-threshold selectivity, and perceptual blur.

Table 6: Simulation parameter grid

Parameter	Grid values	# levels
Speaker architecture	Incremental, global	2
Semantic regime	Context-updating, context-fixed	2
n_{obj}	2, 6, 10, 14, 18, 22, 26, 30	8
σ_{spread}	2.0, 7.75, 15.0	3
ν	0.90, 0.92, 0.94, 0.96, 0.98	5
k	0.10, 0.30, 0.50, 0.70, 0.90	5
w_f	0.1, 0.2, 0.3, 0.5, 0.8	5
Random context samples per combination	1,000	
Total model evaluations	12,000,000	

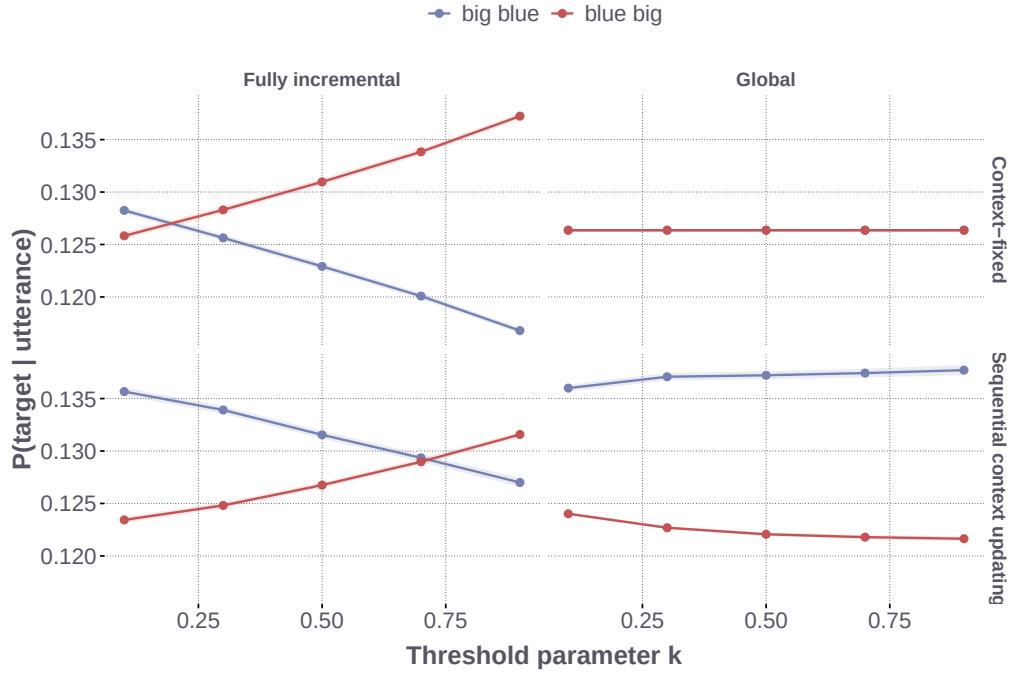


Figure 15: Effect of the threshold parameter k on simulated pragmatic target recovery, separated by speaker and semantic regime. Curves average over the remaining settings; ribbons show ± 1.96 scene-level Monte Carlo standard errors.

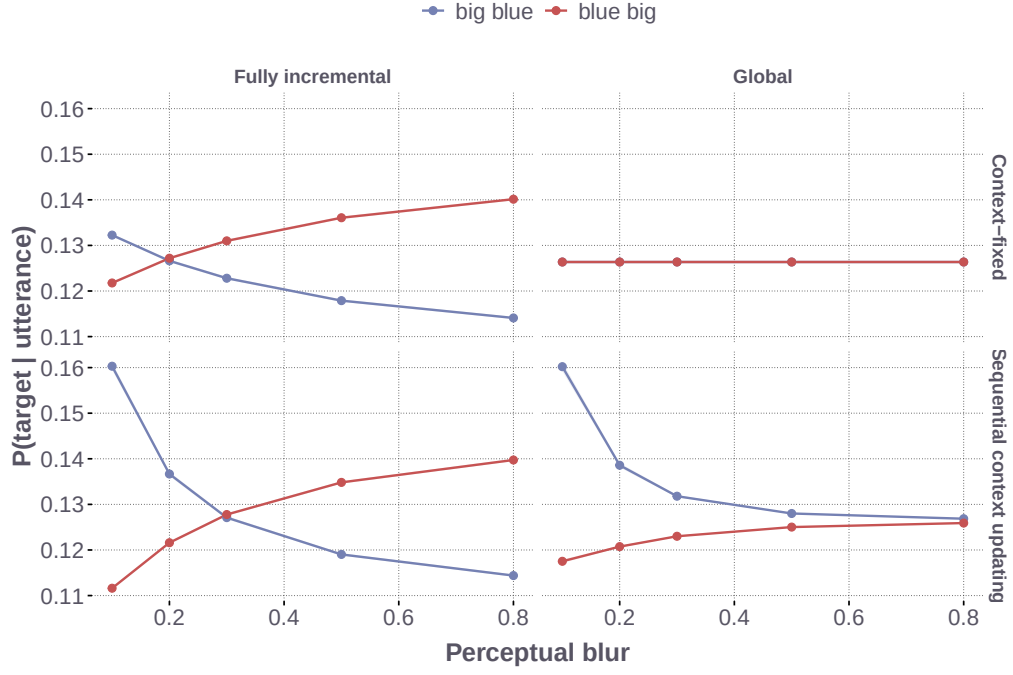


Figure 16: Effect of perceptual blur w_f on simulated pragmatic target recovery, separated by speaker and semantic regime. Curves average over the remaining settings; ribbons show ± 1.96 scene-level Monte Carlo standard errors.

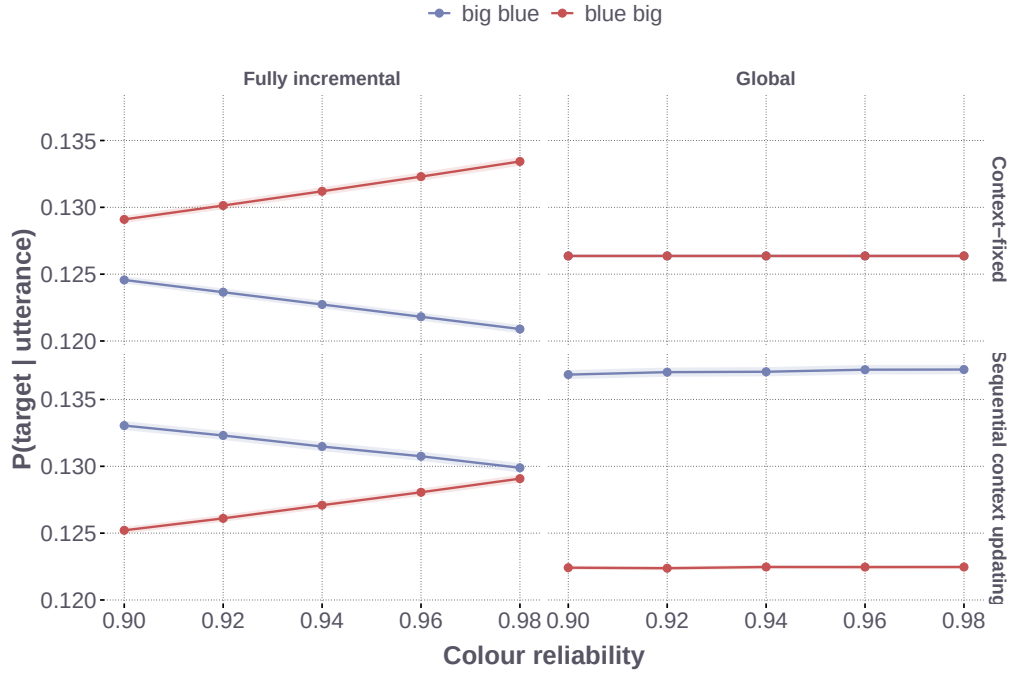


Figure 17: Effect of colour reliability ν on simulated pragmatic target recovery, separated by speaker and semantic regime. Curves average over the remaining settings; ribbons show ± 1.96 scene-level Monte Carlo standard errors.

H.1 Controlled Changes on the Same Random Scenes

The controlled comparisons reuse all 24,000 scenes from the original sweep and the target-recovery measure in Equation (13). Each adjective retains the same extension when the intended referent changes; the referent prior is uniform. The response spaces contain two orders, four size–colour responses including the single adjectives, or the fifteen production responses. For expanded spaces, the pragmatic listener uses each order’s probability under the full speaker distribution, without first conditioning on the two orders. The comparisons cross terminal evaluation with prefix evaluation at $\kappa = 0, .5, 1$, using the original 125 semantic settings or the production constants. Prefix scores and local candidate sets are recomputed for each utterance space. They retain the production convention of no separate STOP action and final normalisation over completed paths. The full fifteen-response implementation agrees with the production forward calculation with task, salience, order, and lapse terms set to zero. This isolates the computational components without assigning fitted participant effects or experimental task labels to random scenes.

The main figure uses the production semantic implementation throughout. As a separate implementation control, substituting its lexical calculation into the original two-order global rule reduces the mean updating advantage from 1.465 to 1.142 percentage points across the original grid. The production terminal score additionally retains its listener log-probability floor, giving the 1.141-point advantage in Figure 12. The implementations differ in probability floors and numerical normalisation around weighted quantiles; the control measures their combined impact. The original curves are retained in Figure 14.

Figure 18 reports the fifteen-response comparison across evaluation rules and local-normalisation weights. The final panel adds the same stable-order coefficient, $\exp(.962357)$, to every model at the production semantic constants. This representative setting comes from the fixed model’s population log-weight location; it does not integrate posterior or participant uncertainty. At $\kappa = 0$, adding this term gives mean advantages of 2.154 and 2.157 percentage points for fixed and updating semantics, respectively. The positive advantages are already present without this term. All other nuisance terms remain zero, and $\alpha = 1$ throughout.

For every contrast, semantic parameter settings are averaged within each scene before calculating Monte Carlo uncertainty. The overall mean gives equal weight to the 24 context-size and size-spread groups. These averages summarise the specified parameter grid; individual contexts can favour different orders. The stored two-order outputs are exactly reproduced, and the controlled forward probabilities agree with the corresponding production calculation to numerical precision.

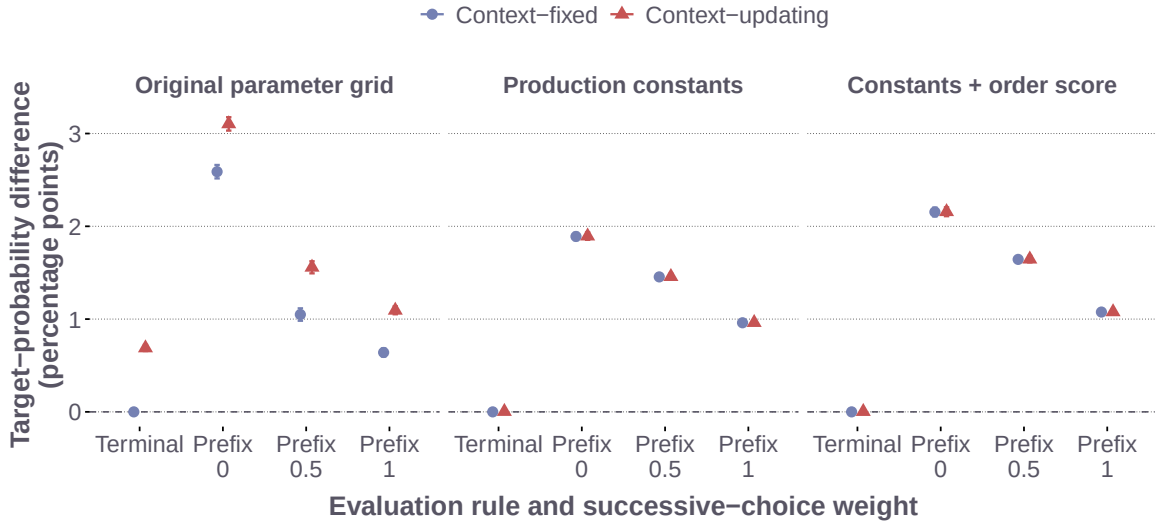


Figure 18: Controlled fifteen-response comparison across evaluation rules, semantic parameters, and a shared stable-order score. Terminal evaluation uses only the completed-expression score. Prefix evaluation accumulates prefix scores with local-normalisation weight $\kappa = 0, .5, 1$. Panels use the original semantic grid, the production semantic constants, and those constants with the stable-order score added. Points and intervals show mean target-probability differences and ± 1.96 scene-level Monte Carlo standard errors. All comparisons use the same random scenes; no participant posterior uncertainty is represented.

H.2 Supplementary Predictions on Experimental Displays

The supplementary simulation uses the 3,034 size–colour trials and all eight fitted production models. For each model, 50 evenly spaced retained draws from each of its four chains give 200 posterior parameter draws. The simulation preserves the fifteen named responses, fitted semantic constants, participant parameters, stable-order score, salience calculation, and task link. For each trial and posterior draw, each of the six objects is evaluated as the intended referent while the adjective extensions and observed task covariates remain fixed. Thus changing the candidate referent does not change which colour or form the uttered words denote. Because the production responses were coded by adjective type, fixing these named adjective extensions across candidate referents is an additional assumption of the listener calculation. This is a model-based extension to counterfactual referents within the observed task context.

Let $S_{it}^{(d)}(u \mid o, X)$ be the resulting complete-response probability at posterior draw d . We evaluate the speaker’s size-first probability conditional on the size–colour adjective set as

$$\frac{S_{it}^{(d)}(\text{SC} \mid o^*, X)}{S_{it}^{(d)}(\text{SC} \mid o^*, X) + S_{it}^{(d)}(\text{CS} \mid o^*, X)}. \quad (27)$$

Separately, the pragmatic listener normalises the full-response speaker probabilities over the six referents using a uniform prior. The listener therefore retains the fitted fifteen-response normalisation when evaluating each order. We average each quantity over the original size–colour trials within posterior draw, then report the posterior mean and central 95% interval. Replaying the original target reproduces its stored likelihood within 10^{-8} for every model.

Figure 19 reports the resulting speaker and listener predictions. In the participant-weight plan-guided models, the mean size-first target-probability advantage was .0062 under fixed semantics, 95% posterior interval [.0047, .0077], and .0058 under updating, [.0045, .0072]. The corresponding mean probabilities of size-first conditional on the size–colour adjective set were .967 and .966. Both fitted regimes therefore generate a size-first advantage under this counterfactual listener calculation. These posterior-derived predictions, together with the response-score decomposition, show how a simulated ordering advantage can coexist with lower predictive accuracy for human adjective content and length.

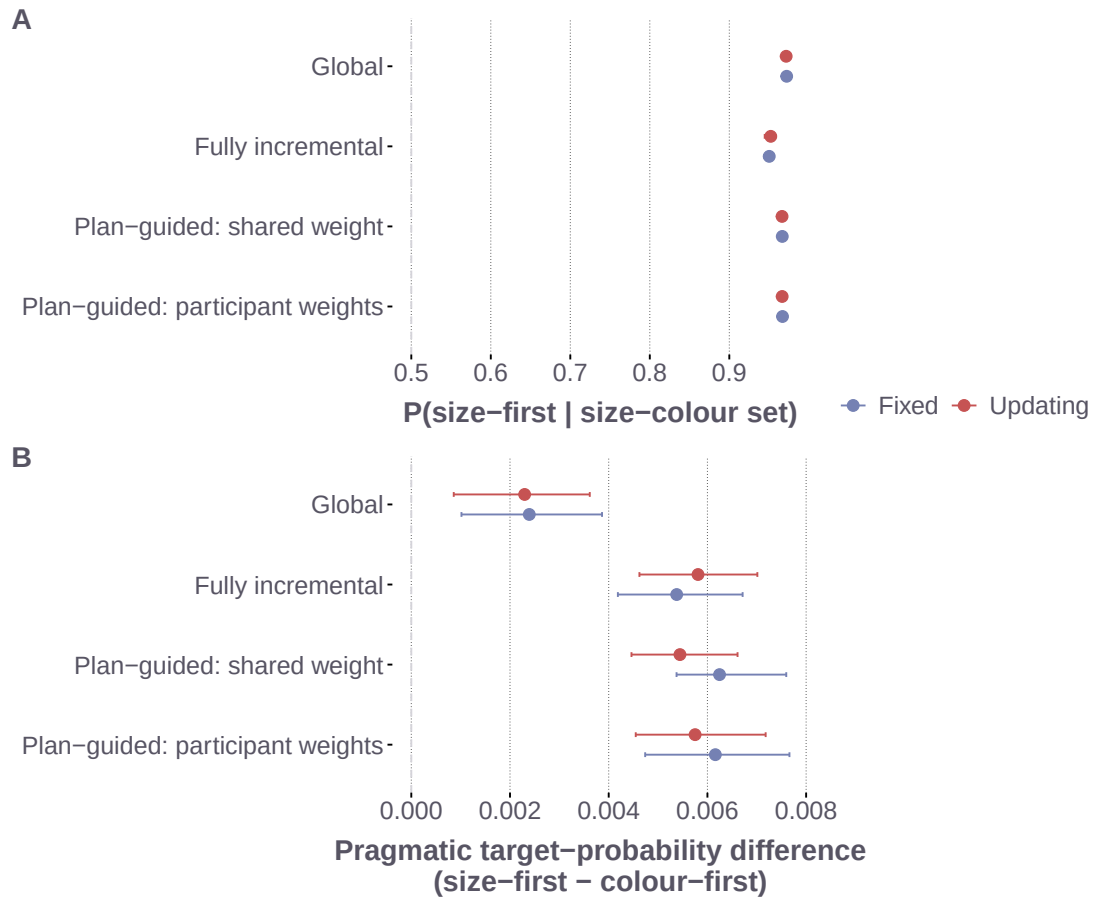


Figure 19: Supplementary predictions from the fitted production models on the size-colour displays. Panel A shows mean speaker probability of size-first conditional on selecting the size-colour adjective set. Panel B shows the difference in pragmatic target probability between size-first and colour-first. Both panels preserve the fitted fifteen-response distribution; the listener in Panel B uses a uniform prior over six referents. Intervals are central 95% posterior intervals over trial-averaged quantities using 200 saved parameter draws per model.